Check for updates

Concise Report

rhodium, intramolecular [4+2] cycloaddition, allenes, 1,3-dienes, chirality transfer, asymmetric synthesis

CJC Chinese Journal of Chemistry 中国化学 · An International Journal

One Stone for Three Birds-Rhodium-Catalyzed Highly Diastereoselective Intramolecular [4+2] Cycloaddition of Optically Active Allene-1,3-dienes

Yulin Han,[a,b] Anni Qin,[c] and Shengming Ma*[,a,c]

[a] *State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, 345 Lingling Lu, Shanghai 200032, China*
[b] *University of Chinese Academy of Sciences, Beijing 100049, China*
[c] *Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, 220 Handan Road, Shanghai 200433, China*

Cite this paper: *Chin. J. Chem.* **2019**, *37*, 486—496. **DOI: 10.1002/cjoc.201900023**

Summary of main observation and conclusion $RhCl(PPh_3)_3$-catalyzed [4+2] intramolecular cycloaddition of optically active axially chiral allene-dienes afforded *cis*-fused [3.4.0]-bicyclic products with three chiral centers in good yields with an excellent chemo- and diastereoselectivity. A pair of enantiomers of such products was generated highly selectively from both enantiomers of starting allene-dienes, indicating that the axial chirality dictated the absolute configurations of the three *in situ* generated chiral centers with a very high efficiency of chirality transfer.

Background and Originality Content

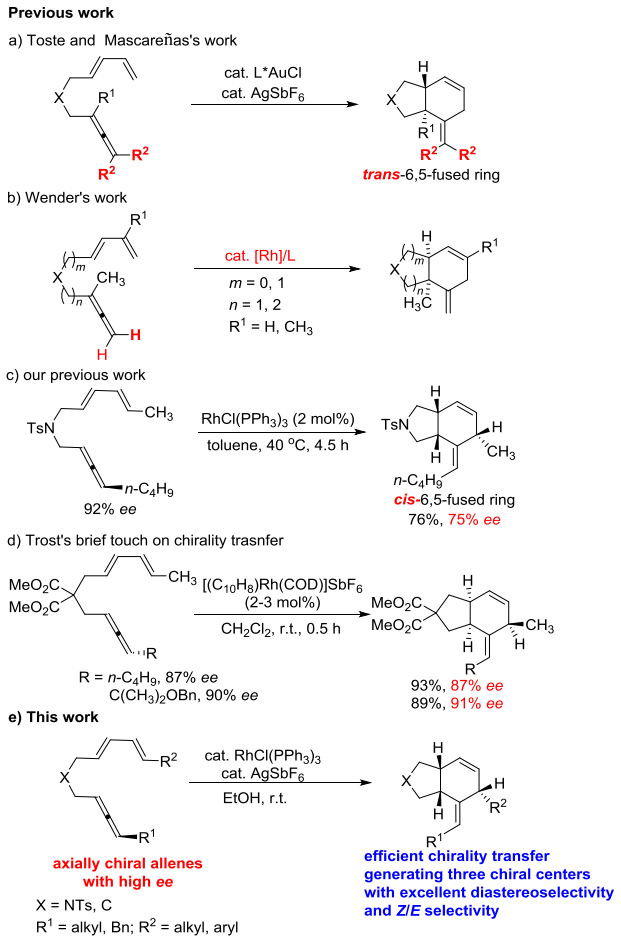
The intramolecular [4+2] cycloaddition between 1,3-dienes and allene may afford bicycles with three chiral centers in one step.[1] In 2009, the groups of Toste and Mascareñas independently demonstrated the gold-catalyzed intramolecular [4+2] cycloaddition reaction of allene-dienes which have two same R^2 groups at the "distal" terminal of the allene to give ***trans***-6,5-fused rings (Scheme 1, a).[2] Wender (Scheme 1, b)[3] and we (Scheme 1, c)[4] reported the rhodium-catalyzed intramolecular [4+2] cycloaddition of 1,3-disubstituted allene-dienes to afford ***cis***-6,5-fused rings. However, the [4+2] cycloadditions of such substrates with an optically active allene unit afforded the products in low *ee* albeit with excellent de.[4] Trost *et al.* had a very brief study on this topic using substrates of 90% *ee* (Scheme 1, d), by using the cationic catalyst $[(C_{10}H_8)Rh(COD)]SbF_6$,[5] which would not work for the reaction of substrates containing NTs as the linker.[4] Thus, it is of high interest to develop a general and efficient cycloaddition of chiral allene-dienes. Herein, we present our new observation on the Rh(I)-catalyzed intramolecular [4+2] cycloaddition of optically active disubstituted allene-dienes, in which the internal C=C bond of the allene moiety reacted with the 1,3-dienes to produce *cis*-6,5-fused bicyclic products with an excellent *Z*/*E* selectivity for the remaining C=C bond with an excellent diastereoselectivity and efficiency of chirality transfer affording the bicyclic products in high *ee* (Scheme 1, e).

Results and Discussion

Synthesis of starting materials

Synthesis of optically active allenes 1.[6] Optically active allenes (R_a)-**1a**—**1d** were prepared *via* the $CuBr_2$-catalyzed EATA (**e**nantioselective **a**llenation of **t**erminal **a**lkynes) reaction from propargyl tosylamide or dimethyl propargylmal-onate, (*S*)-2-(diphenylhydroxymethyl) pyrrolidine and commercially available aldehydes in 47% to 71% yields with ≥ 92% *ee* (Eq. 1, Scheme 2). (S_a)-**1b**—**1c** were obtained in 54%—70% yields with ≥ 93% *ee* with the aid of (*R*)-2-(diphenylhydroxymethyl)pyrrolidine (Eq. 2, Scheme 2).

Scheme 1 Metal-catalyzed intramolecular [4+2] cycloaddition of allene-dienes

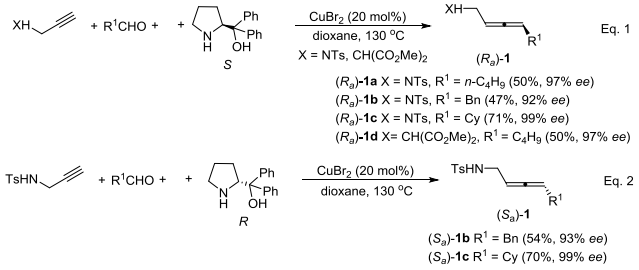


*E-mail: masm@sioc.ac.cn; Fax: 0086-021-64167510

For submission: https://mc.manuscriptcentral.com/cjoc
For articles: https://onlinelibrary.wiley.com/journal/16147065

© 2019 SIOC, CAS, Shanghai, & WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

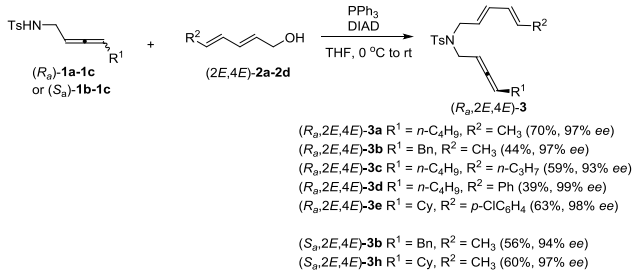
Scheme 2 The synthesis of optically active allenes-EATA reaction



Synthesis of (2*E*,4*E*)-dienol. (2*E*,4*E*)-Hexa-2,4-dien-1-ol (2*E*,4*E*)-**2a** was obtained in 58% yield using the reduction of sorbic acid with $LiAlH_4$.[7] Dien-1-ol compounds **2b**,[8] **2c**[9] and **2d**[10] were synthesized in two steps from α,β-unsaturated enals: ethyl (2*E*,4*E*)-alkadienoate compounds were obtained by the Horner–Wadsworth–Emmons reaction of α,β-unsaturated aldehydes, followed by the reduction with DIBAL-H to afford (2*E*,4*E*)-alkadien-1-ol compounds **2b**—**2d** in 71% to 86% yields.

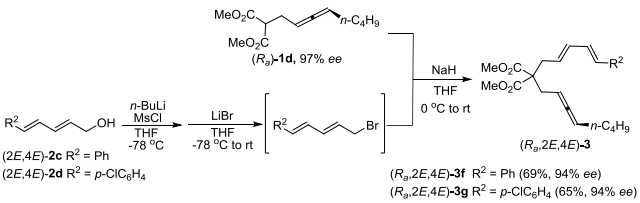
Synthesis of optically active allene-dienes (R_a/S_a,2*E*,4*E*)-3a—3e with NTs tether.[11] 1,3-Disubstituted allene-dienes (R_a,2*E*,4*E*)-**3a**—**3e** were prepared *via* a Mitsunobu reaction of the corresponding allenes **1a**—**1c** and (2*E*,4*E*)-dien-1-ol compounds **2a**—**2d** in 39%—70% yields with 93%—99% *ee* (Scheme 3). (S_a,2*E*,4*E*)-**3b** and (S_a,2*E*,4*E*)-**3h** were obtained in 56% and 60% yields with 94% and 97% *ee*, respectively.

Scheme 3 Synthesis of allene-dienes (R_a,2*E*,4*E*)-**3** with NTs tether



Synthesis of optically active allene-dienes (R_a,2*E*,4*E*)-3f and (R_a,2*E*,4*E*)-3g with malonate tether.[2d] (R_a,2*E*,4*E*)-**3f** and (R_a,2*E*,4*E*)-**3g** were synthesized by the alkadienylation of sodium salts of (R_a)-**1d** with (2*E*,4*E*)-2,4-alkadienyl bromides, which were prepared *via* the reaction of dienols (2*E*,4*E*)-**2c** and (2*E*,4*E*)-**2d** with methanesulfonyl chloride and LiBr (Scheme 4).

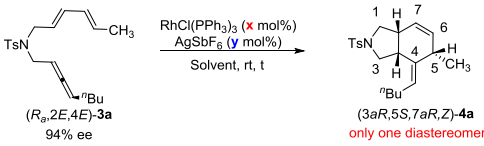
Scheme 4 Synthesis of allene-dienes (R_a,2*E*,4*E*)-**3f** and (R_a,2*E*,4*E*)-**3g** with malonate tether



Rh-catalyzed highly diastereoselective [4+2] cycloaddition reaction of axially chiral allene-1,3-dienes

When the reaction of optically active 1,3-diene-allene (*R*a)-**3a** was conducted with $RhCl(PPh_3)_3$ (2 mol%) in toluene at 40 °C, although only one diastereomer (3a*R*,5*S*,7a*R*,*Z*)-**4a** was produced in 76% isolated yield, the enantiopurity of the product was rather disappointing-only 75% *ee* (entry 1, Table 1).[4] There was a slight increase in the yield and enantiopurity when running the reaction at room temperature (entry 2, Table 1). It was observed that the polar solvents such as 1,4-dioxane, THF, dichloromethane, and chloroform all failed to increase the yield and enantiopurity and the reaction was slower (entries 3—6, Table 1). The enantiopurity was improved to 87% with the addition of 3 mol% $AgSbF_6$, albeit the reaction became very sluggish (entry 7, Table 1). To our delight, (3a*R*,5*S*,7a*R*,*Z*)-**4a** was produced with 93% *ee* and 60% isolated yield when the reaction of (*R*a)-**3a** was conducted with $RhCl(PPh_3)_3$ (3 mol%) and $AgSbF_6$ (5 mol%) in toluene (entry 8, Table 1). Using ethanol as the solvent improved the yield further up to 87% but the product was only with 76% *ee* in the absence of $AgSbF_6$ (entry 9, Table 1). We reasoned that the coordination ability of EtOH is helping, thus, $AgSbF_6$ (5 mol%) was then added again to make a cationic Rh catalyst. Indeed, a complete chirality transfer and excellent yield were obtained, indicating the importance of a cationic nature of the Rh catalyst (entry 10, Table 1). The commercial absolute ethanol was also fine with a lower yield (entry 11, Table 1). Thus, the standard conditions were defined as conducting the reaction with $RhCl(PPh_3)_3$ (3 mol%) and $AgSbF_6$ (5 mol%) in anhydrous ethanol.

Table 1 Rhodium-catalyzed intramolecuar [4+2] cycloaddition of (R_a)-1,3-disubstituted allene-diene (R_a)-**3a**[a]



Entry	*x*	*y*	Solvent	*t*/h	(3a*R*,5*S*,7a*R*,*Z*)-**3a**	
					Isolated yield/%	*ee*/%
1[b]	2	–	Toluene	4.5	76	75
2	2	–	Toluene	10	81	78
3[c]	2	–	Dioxane	58	37	74
4[c]	2	–	THF	58	31	74
5	2	–	DCM	14	73	77
6	2	–	$CHCl_3$	15.5	67	70
7[c]	2	3	Toluene	108	46	87
8	3	5	Toluene	8	60	93
9	3	–	**EtOH**	3.5	87	76
10[d]	3	5	**EtOH**	3.5	74	94
11[e]	3	5	EtOH	6	55	93

[a] The reaction was conducted with (R_a,2*E*,4*E*)-**3a** (0.2 mmol), $RhCl(PPh_3)_3$, and $AgSbF_6$ in 2 mL of solvent. [b] The reaction was conducted at 40 °C. [c] The recoveries of (R_a,2*E*,4*E*)-**3a** of entries 3, 4, and 7 were 22%, 20%, and 26%, respectively. [d] (R_a,2*E*,4*E*)-**3a** with 95% *ee*. [e] The commercial absolute ethanol was not dried with 4 Å molecular sieves and used directly.

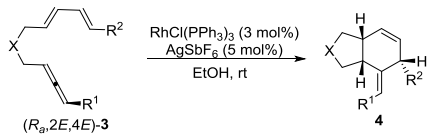
With the optimal reaction conditions in hand, the scope of the reaction was investigated. Firstly, we studied the substrates with NTs as the linker. With R^2 being methyl, *n*-propyl, phenyl, and *p*-ClC_6H_4, the reaction afforded the corresponding product (3a*R*,5*S*,7a*R*,*Z*)-**4a**—**4e** in moderate to excellent yields with a very high efficiency of axial chirality transfer generating three chiral centers (entries 1—5, Table 2). The absolute configuration of the product was established by the X-ray single crystal diffraction study of (3a*R*,5*S*,7a*R*,*Z*)-**4b** (Figure 1).[12] Malonate-tether (3a*R*,5*R*,7a*R*,*E*)-**4f** and (3a*R*,5*R*,7a*R*,*E*)-**4g** were formed when R^2 was Ph or *p*-ClC_6H_4 with perfect efficiency of chirality transfer and excellent yields (entries 6 and 7, Table 2).

Based on the absolute configuration of the starting material, the X-ray diffraction study of (3a*R*,5*S*,7a*R*,*Z*)-**4b**, and the observed diastereoselectivity, a mechanism involving cyclometalation for the rhodium-catalyzed intramolecular [4+2] cycloaddition of allene-dienes has been proposed in our previous study (Scheme 5).

 © 2019 SIOC, CAS, Shanghai, & WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

16147065, 2019, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.201900023 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2 The scope of rhodium-catalyzed intramolecular [4+2] cycloaddition of (R_a)-1,3-disubstituted allene-diene[a]



Entry	(R_a,2*E*,4*E*)-**3** (*ee*/%) X/R^1/R^2	*t*/h	**4** Isolated yield/%	*ee*/%
1	NTs/^{n}Bu/CH_3 (**3a**) (97)	2	73 ((3a*R*,5*S*,7a*R*,*Z*)-**4a**)	97
2	NTs/Bn/CH_3 (**3b**) (97)	9	62 ((3a*R*,5*S*,7a*R*,*Z*)-**4b**)	97
3	NTs/^{n}Bu/C_3H_7 (**3c**) (93)	9	75 ((3a*R*,5*S*,7a*R*,*Z*)-**4c**)	93
4[b]	NTs/^{n}Bu/Ph (**3d**) (98)	5	59 ((3a*R*,5*R*,7a*R*,*Z*)-**4d**)	96
5	NTs/Cy/*p*-ClC_6H_4 (**3e**) (98)	10	79 ((3a*R*,5*R*,7a*R*,*Z*)-**4e**)	99
6	$C(CO_2Me)_2$/^{n}Bu/Ph (**3f**) (94)	23	82 ((3a*R*,5*R*,7a*R*,*E*)-**4f**)	94
7	$C(CO_2Me)_2$/^{n}Bu/*p*-ClC_6H_4 (**3g**) (94)	9	80 ((3a*R*,5*R*,7a*R*,*E*)-**4g**)	94

[a] The reaction was conducted with (*R*a,2*E*,4*E*)-**3** (0.5 mmol), $RhCl(PPh_3)_3$ (3 mol%), and $AgSbF_6$ (5 mol%) in 5 mL of dry EtOH at rt. [b] The reaction was conducted at 40 °C.

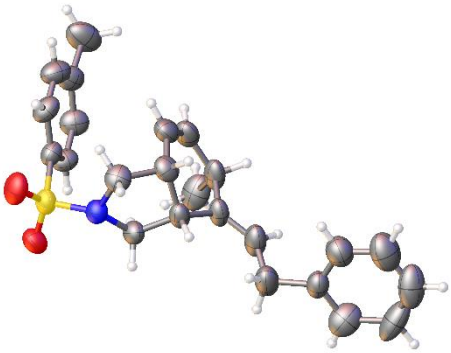
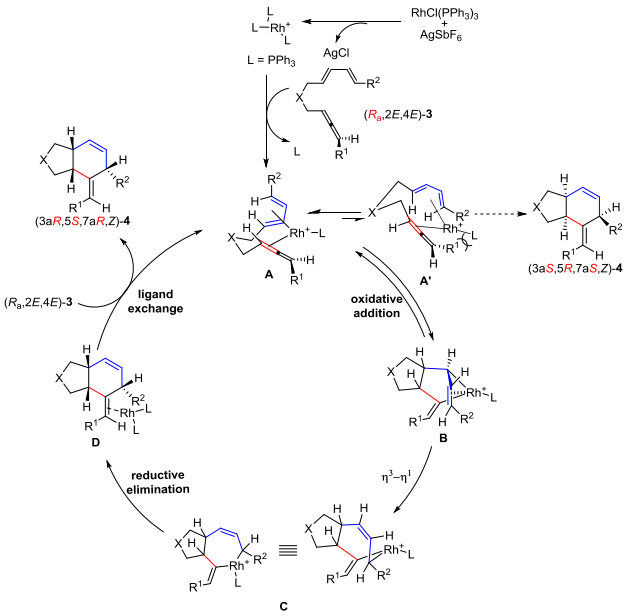


Figure 1 The ORTEP representation of (3a*R*,5*S*,7a*R*,*Z*)-**4b**.

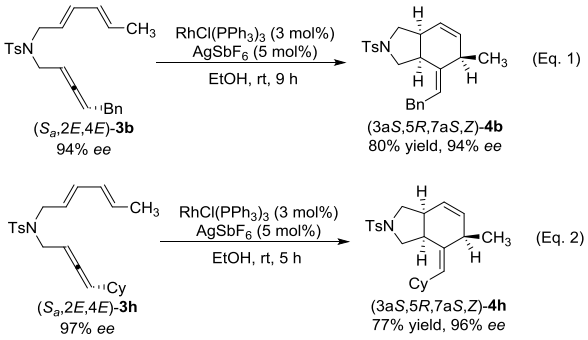
R-Isomer (R_a,2*E*,4*E*)-**3** exchanged with ligand of the cationic Rh catalyst formed by the reaction of $RhCl(PPh_3)_3$ with $AgSbF_6$ would generate Intermediate **A** and **A'**. In order to make the R^1 group pointing away from the 1,3-diene unit, the "inner" C=C bonds in allene and the *s-cis*-diene coordinated with the rhodium atom with the two H atoms in intermediate **A** and **A'** *cis* orientated. The intermediate **A** is favored since the steric hindrance between R^1 and the catalyst in intermediate **A'** is much larger. Subsequent cyclometalation and allylic rearrangement lead to the formation of rhodiabicycloheptene **C**, in which the three tertiary hydrogen atoms are *cis* orientated. Subsequent reductive elimination and ligand exchange with (R_a,2*E*,4*E*)-**3** would yield (3a*R*,5*S*,7a*R*,*Z*)-**4** and regenerate the catalytically active specie **A** to finish the catalytic cycle.[13]

If such a rationale is indeed working, the reaction of (S_a,2*E*,4*E*)-**3b** and (S_a,2*E*,4*E*)-**3h** should afford the enantiomeric products (3a*S*,5*R*,7a*S*,*Z*)-**4b** and (3a*S*,5*R*,7a*S*,*Z*)-**4h**. Indeed, this is the case, (3a*S*,5*R*,7a*S*,*Z*)-**4b** and (3a*S*,5*R*,7a*S*,*Z*)-**4h** were formed in 80% yield with 94% *ee* and 77% yield with 96% *ee*, respectively (Eqs. 1 and 2, Scheme 6). The absolute configuration of (3a*S*,5*R*,7a*S*,*Z*)-**4b** was established by the X-ray single crystal diffraction (Figure 2).[14] In conclusion, the absolute configurations of the C3a, C5, and C7a in the cycloaddition product are controlled selectively by the axial chirality of the allene moiety of the substrate.

Scheme 5 A proposed mechanism for the intramolecular [4+2] cycloaddition of allene-dienes (*R*a,2*E*,4*E*)-**3**



Scheme 6 Rhodium-catalyzed intramolecuar [4+2] cycloaddition of (S_a,2*E*,4*E*)-**3b** and (S_a,2*E*,4*E*)-**3h**



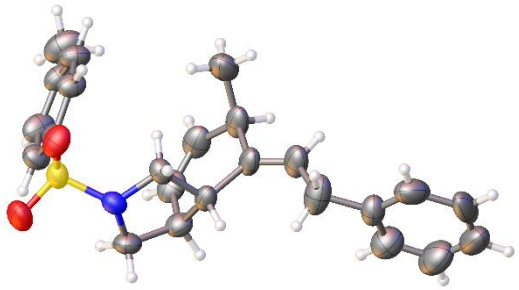


Figure 2 The ORTEP representation of (3a*S*,5*R*,7a*S*,*Z*)-**4b**.

Conclusions

We have developed the $RhCl(PPh_3)_3$/$AgSbF_6$-cocatalyzed intramolecular [4+2] cycloaddition of optically active axially chiral 1,3-disubstituted allene-dienes, *cis*-6,5-fused bicyclic products

 © 2019 SIOC, CAS, Shanghai, & WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

16147065, 2019, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.201900023 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

with three chiral centers could be obtained with an excellent diastereoselectivity and efficient chirality transfer. In addition, the absolute configurations in the C3a, C5, and C7a of the products are controlled by the absolute configuration of the axial allene chirality. A meachanism involving concerted cyclometalation, allylic rearrangement, and reductive elimination mechanism accounts for the observed efficiency for chirality transfer and diastereoselectivity. Further studies including the catalyzed enantioselective [4+2] cycloaddition of racemic allene-dienes substrates by applying chiral ligand are being conducted in our laboratory.

Experimental

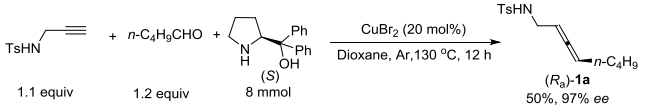
General information

NMR spectra were taken with an Agilent 400 MR DD2 or Varian 400-MR spectrometer (400 MHz for ^{1}H NMR and 100 MHz for ^{13}C NMR) in $CDCl_3$. NMR spectra were taken using TMS (^{1}H, δ = 0), residual $CHCl_3$ (δ 7.26) in $CDCl_3$ as the internal standards, respectively. Chemical shifts were recorded in ppm and coupling constants were reported in Hz. Mass and HRMS spectra were carried out in EI or ESI mode. All reactions have been carried out in oven-dried Schlenk tubes. $RhCl(PPh_3)_3$ (99%) was purchased from J&k and TCI, $AgSbF_6$ (99%) was purchased from Alfa Aesar; 4 Å molecular sieves were purchased from Alfa Aesar and kept in glove box after activation (heated at 450 °C for 10 h in Muffle furnace, taken out after cooling to 200 °C and then kept in glove box to allow to cool to room temperature). Petroleum ether (boiling range: 60—90 °C) was purchased from Adamas. Toluene, 1,4-dioxane, and THF were dried over sodium wire with benzophenone as the indicator and distilled freshly before use. CH_2Cl_2 and $CHCl_3$ were dried over CaH_2 and distilled freshly before use. The commercial absolute ethanol (EtOH) was dried with 4 Å molecular sieves. Other reagents were used as received without further treatment. All the temperatures are referred to the oil baths used.

Synthesis of starting materials

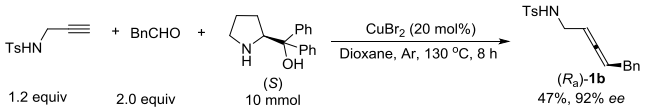
Synthesis of optically active 1,3-disubstituated allene

(1) Preparation of (R_a)-*N*-(2,3-octadien-1-yl) toluenesulfonamide(R_a)-1a (hanyl-9-060)[6]



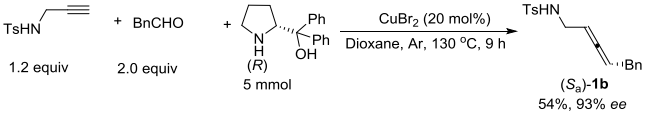
To a flame-dried three neck flask with a reflux condenser, (*S*)-2-(diphenylhydroxymethyl) pyrrolidine (2.0703 g, 8 mmol), $CuBr_2$ (0.3577 g, 1.6 mmol, 20 mol%), propargyl tosylamide (1.8387 g, 8.8 mmol, 1.1 equiv)/dioxane (10 mL), and pentanal (0.8387 g, 9.6 mmol, 1.2 equiv)/dioxane (20 mL) were sequentially added under argon atmosphere. The reaction was complete after being stirred in an oil bath preheated at 130 °C for 12 h, which was monitored by TLC. After cooling to room temperature, the resulting mixture was diluted with diethyl ether (50 mL) and washed with an aqueous solution of hydrochloric acid (3 mol·L^{-1}, 50 mL). The organic layer was separated, and the aqueous layer was extracted with ether (50 mL × 2). The combined organic layer was washed with 50 mL of brine and dried over anhydrous $MgSO_4$. After filtration and evaporation, the residue was purified by chromatography on silica gel to afford (R_a)-**1a** (1.1188 g, 50%) (eluent: petroleum ether/ethyl acetate = 8/1) as a liquid: 97% *ee* (HPLC conditions: chiralcel AD-H column, *n*-hexane/*i*-PrOH = 98/2, 1.5 mL/min, λ = 214 nm, t_R (major) = 29.02 min, t_R (minor) = 33.09 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.77 (d, *J* = 8.0 Hz, 2H, Ar-H), 7.30 (d, *J* = 8.4 Hz, 2H, Ar-H), 5.21—5.13 (m, 1H, HC=), 5.06—4.98 (m, 1 H, HC=), 4.94 (t, *J* = 5.8 Hz, 1H, NH), 3.55 (td, J_1 = 6.0 Hz, J_2 = 2.8 Hz, 2H, NCH_2), 2.42 (s, 3H, CH_3), 1.97—1.87 (m, 2H, CH_2), 1.35—1.25 (m, 4H, 2 × CH_2), 0.87 (t, *J* = 7.0 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 203.3, 143.3, 136.9, 129.5, 127.0, 94.3, 87.5, 41.9, 30.9, 28.0, 22.0, 21.4, 13.7; MS (ESI) *m*/*z*: 302 ([M+Na]$^+$), 280 ([M+H]$^+$); IR (neat) *v*: 3277, 2956, 2926, 2858, 1964, 1598, 1495, 1418, 1323, 1185, 1156, 1092, 1071 cm^{-1}; HRMS calcd for $C_{15}H_{21}NO_2S$ (M$^+$): 279.1293, found: 279.1300.

(2) Preparation of (R_a)-*N*-(5-phenyl-2,3-pentadien-1-yl) toluene-sulfonamide (R_a)-1d (hanyl-10-022)[6]



Typical procedure A: To a flame-dried sealed tube were added (*S*)-2-(diphenylhydroxymethyl) pyrrolidine (2.5861 g, 10 mmol), $CuBr_2$ (0.4516 g, 2.0 mmol, 20 mol%)/(10 mL), benzaldehyde (2.5302 g, 20 mmol, 2.0 equiv)/dioxane (10 mL), and propargyl tosylamide (2.5110 g, 12 mmol, 1.2 equiv)/dioxane (10 mL) sequentially under argon atmosphere. The tube was then sealed by screwing the polytetrafluoroethylene plug tightly. The reaction was complete after being stirred in an oil bath preheated at 130 °C for 8 h, which was monitored by TLC. After cooling to room temperature, the resulting mixture was diluted with ether (50 mL) and washed with an aqueous solution of hydrochloric acid (3 mol·L^{-1}, 50 mL). The organic layer was separated, and the aqueous layer was extracted with ether (50 mL × 2). The combined organic layer was washed with brine (50 mL) and dried over anhydrous $MgSO_4$. After filtration and evaporation, the residue was purified by chromatography on silica gel to afford (R_a)-**1b**[15] (1.4609 g, 47%) (eluent: petroleum ether/ethyl acetate = 8/1) as a liquid: 92% *ee* (HPLC conditions: Chiralcel OD-H column, *n*-hexane/*i*-PrOH = 80/20, 1.0 mL/min, λ = 214 nm, t_R(major) = 21.48 min, t_R(minor) = 33.11 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.72 (d, *J* = 8.0 Hz, 2 H, Ar-H), 7.32—7.18 (m, 5H, Ar-H), 7.15 (d, *J* = 7.6 Hz, 2H, Ar-H), 5.41—5.32 (m, 1H, HC=), 5.06—4.97 (m, 1H, HC=), 4.48 (t, *J* = 5.8 Hz, 1H, NH), 3.58—3.42 (m, 2H, NCH_2), 3.26 (dd, J_1 = 6.8 Hz, J_2 = 2.4 Hz, 2H, CH_2), 2.41 (s, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 204.2, 143.4, 139.5, 136.9, 129.6, 128.4, 128.38, 127.1, 126.3, 94.0, 88.4, 41.9, 34.9, 21.5; MS (ESI) *m*/*z*: 336 ([M+Na]$^+$), 314 ([M+H]$^+$); IR (neat) *v*: 3278, 3061, 3027, 2920, 2867, 1965, 1598, 1494, 1453, 1424, 1323, 1156, 1093, 1072, 1031 cm^{-1}.

(3) Preparation of (S_a)-*N*-(5-phenyl-2,3-pentadien-1-yl) toluenesulfonamide (S_a)-1b (hanyl-10-007)

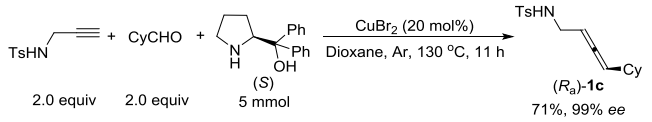


Following **Typical procedure A**, the reaction of (*R*)-2-(diphenylhydroxymethyl) pyrrolidine (1.2941 g, 5 mmol), $CuBr_2$ (0.2263 g, 1.0 mmol, 20 mol%), benzaldehyde (1.2050 g, 10 mmol, 2.0 equiv)/dioxane (10 mL), and propargyl tosylamide (1.2542 g, 6 mmol, 1.2 equiv)/dioxane (5 mL) at 130 °C for 9 h afforded (S_a)-**1b** (0.8489 g, 54%) (eluent: petroleum ether/ethyl acetate = 8/1) as a liquid: 93% *ee* (HPLC conditions: chiralcel OD-H column, *n*-hexane/*i*-PrOH = 80/20, 1.0 mL/min, λ = 214 nm, t_R(minor) = 21.60 min, t_R(major) = 31.14 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.72 (d, *J* = 8.0 Hz, 2 H, Ar-H), 7.35—7.19 (m, 5H, Ar-H), 7.15 (d, *J* = 7.2 Hz, 2 H, Ar-H), 5.43—5.32 (m, 1H, HC=), 5.09—4.98 (m, 1H, HC=), 4.44 (t, *J* = 5.4 Hz, 1H, NH), 3.57—3.42 (m, 2 H, NCH_2), 3.27 (dd, J_1 = 6.8 Hz, J_2 = 2.0 Hz, 2H, CH_2), 2.41 (s, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 204.3, 143.4, 139.6, 136.9, 129.6, 128.5, 128.4, 127.1, 126.3, 94.0, 88.4, 41.9, 34.9, 21.5; MS (ESI) *m*/*z*: 336 ([M+Na]$^+$), 314 ([M+H]$^+$); IR (neat) *v*: 3282, 1966, 1599, 1494, 1453, 1418, 1323, 1264, 1184, 1156, 1120, 1092, 1074, 1031 cm^{-1}; HRMS calcd for $C_{18}H_{23}N_2O_2S$ ([M+NH$_4$]$^+$): 331.1475, found: 331.1471.

 © 2019 SIOC, CAS, Shanghai, & WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

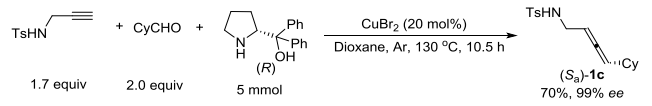
16147065, 2019, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.201900023 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

(4) Preparation of (R_a)-*N*-(4-cyclohexyl-2,3-butadien-1-yl) toluenesulfonamide (R_a)-1c (hanyl-11-006)[6]



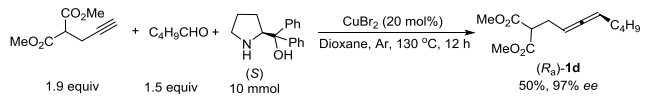
Following **Typical procedure A**, the reaction of (*S*)-2-(diphenylhydroxymethyl) pyrrolidine (1.2933 g, 5 mmol), $CuBr_2$ (0.2265 g, 1.0 mmol, 20 mol%)/dioxane (6 mL), cyclohexylformaldehyde (1.1170 g, 10.0 mmol, 2.0 equiv)/dioxane (2 mL), and propargyl tosylamide (2.0984 g, 10 mmol, 2.0 equiv)/dioxane (2 mL) at 130 °C for 11 h afforded (R_a)-**1c**[16] (1.0850 g, 71%) (eluent: petroleum ether/ethyl acetate = 10/1) as a liquid: 99% *ee* (HPLC conditions: chiralcel AD-H column, *n*-hexane/*i*-PrOH = 98/2, 1.0 mL/min, λ = 214 nm, t_R(major) = 56.23 min, t_R(minor) = 62.33 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.76 (d, J = 8.0 Hz, 2H, Ar-H), 7.30 (d, J = 7.6 Hz, 2H, Ar-H), 5.23—5.16 (m, 1H, HC=), 5.12—5.04 (m, 1H, HC=),4.67 (*t*, J = 4.8 Hz, 1H, NH), 3.62—3.50 (m, 2H, NCH_2), 2.43 (s, 3H, CH_3), 1.98—1.86 (m, 1H, CH), 1.73—1.58 (m, 5H, 2 × CH_2 + one proton of CH_2), 1.30—0.90 (m, 5H, 2 × CH_2 + one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 202.2, 143.3, 137.0, 129.6, 127.1, 100.4, 88.4, 42.0, 36.8, 32.8, 32.7, 25.9, 25.8, 21.4; MS (ESI) *m*/*z*: 328 ([M+Na]$^+$); IR (neat) *v*: 3291, 2923, 2851, 1957, 1595, 1493, 1451, 1422, 1386, 1325, 1290, 1228, 1184, 1154, 1090, 1066, 1040, 1020 cm^{-1}.

(5) Preparation of (S_a)-*N*-(4-cyclohexyl-2,3-butadien-1-yl) toluenesulfonamide (S_a)-1c (hanyl-11-007)



Following **Typical procedure A**, the reaction of (*R*)-2-(diphenylhydroxymethyl) pyrrolidine (1.2919 g, 5 mmol), $CuBr_2$ (0.2263 g, 1.0 mmol, 20 mol%)/dioxane (6 mL), cyclohexylformaldehyde (1.1240 g, 10 mmol, 2.0 equiv)/dioxane (2 mL), and propargyl tosylamide (1.7540 g, 8.4 mmol, 1.7 equiv)/dioxane (2 mL) at 130 °C for 10.5 h afforded (S_a)-**1c** (1.0750 g, 70%) (eluent: petroleum ether/ethyl acetate = 10/1) as a liquid: 99% *ee* (HPLC conditions: Chiralcel AD-H column, *n*-hexane/*i*-PrOH = 98/2, 1.0 mL/min, λ = 214 nm, t_R(minor) = 56.16 min, t_R(major) = 58.68 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.75 (d, J = 8.0 Hz, 2H, Ar-H), 7.31 (d, J = 8.0 Hz, 2H, Ar-H), 5.22—5.17 (m, 1H, HC=), 5.12—5.04 (m, 1H, HC=), 4.58 (t, J = 5.2 Hz, 1H, NH), 3.55 (dd, J_1 = 8.8 Hz, J_2 = 5.6 Hz, 2H, NCH_2), 2.43 (s, 3H, CH_3), 1.98—1.85 (m, 1H, CH), 1.73—1.57 (m, 5H, 2 × CH_2 + one proton of CH_2), 1.32—0.85 (m, 5H, 2 × CH_2 + one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 02.2, 143.3, 137.0, 129.6, 127.1, 100.5, 88.5, 42.0, 36.8, 32.9, 32.7, 25.9, 25.8, 21.4; MS (EI) *m*/*z* (%): 305 (M$^+$, 2.65), 155 (100); IR (neat) *v*: 3290, 2922, 2851, 1958, 1595, 1493, 1451, 1422, 1386, 1325, 1290, 1228, 1184, 1154, 1090, 1066, 1040, 1019 cm^{-1}; HRMS calcd for $C_{17}H_{23}NO_2S$ (M$^+$): 305.1449, found: 305.1440.

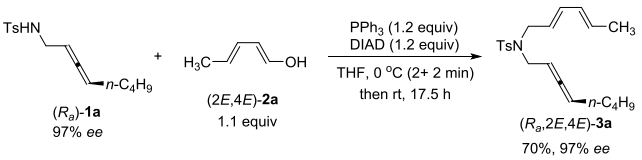
(6) Preparation of (R_a)-dimethyl 2-(2,3-octadien-1-yl) malonate (R_a)-1d (hanyl-9-167)



Following **Typical procedure A**, the reaction of (*S*)-2-(diphenylhydroxymethyl) pyrrolidine (2.5859 g, 10 mmol), $CuBr_2$ (0.4521 g, 2.0 mmol, 20 mol%)/dioxane (10 mL), pentaldehyde (1.2937 g, 15.0 mmol, 1.5 equiv)/dioxane (10 mL), and dimethyl 2-propargylmalonate (3.1862 g, 18.7 mmol, 1.9 equiv) /dioxane (10 mL) at 130 °C for 12 h afforded (S_a)-**1d** (1.2102 g, 50%) (eluent: petroleum ether/ethyl acetate = 30/1) as a liquid: 97% *ee* (HPLC conditions: chiralcel OD-H column, *n*-hexane/*i*-PrOH = 100/1, 0.7 mL/min, λ = 214 nm, t_R(major) = 12.12 min, t_R(minor) = 12.84 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 5.20—5.03 (m, 2H, 2 × HC=), 3.74 (s, 6H, 2 × CH_3), 3.51 (t, J = 7.6 Hz, 1H, CH), 2.63—2.54 (m, 2H, CH_2), 2.00—1.90 (m, 2H, CH_2), 1.41—1.27 (m, 4H, 2 × CH_2), 0.90 (t, J = 7.0 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 203.9, 169.4, 169.3, 92.9, 87.3, 52.5, 51.2, 31.1, 28.4, 28.0, 22.1, 13.8; MS (EI) *m*/*z* (%): 240 (M$^+$, 10.25), 79 (100); IR (neat) *v*: 2927, 2852, 1732, 1620, 1437, 1341, 1233, 1198, 1153, 1032 cm^{-1}; HRMS calcd for $C_{13}H_{20}O_4$ (M$^+$): 240.1362, found: 240.1371.

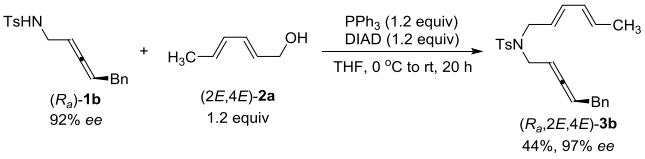
Synthesis of axially chiral allene-1,3-dienes

(1) Preparation of (R_a)-*N*-(2,3-octadien-1-yl)-*N*-[(2*E*,4*E*)-hexadien-1-yl] toluenesulfonamide (R_a,2*E*,4*E*)-3a (hanyl-8-083)



To a dry round-bottom flask placed in an ice bath were added PPh_3 (1.6457 g, 6.24 mmol, 1.2 equiv), (2*E*,4*E*)-hexa-2,4-dien-1-ol **2a** (0.5637 g, 5.72 mmol, 1.1 equiv), (R_a)-*N*-(2,3-octadien-1-yl) toluenesulfonamide (R_a)-**1a** (97% *ee*, 1.4549 g, 5.2 mmol), and 10 mL of THF at 0 °C. Diisopropylazodicarboxylate (DIAD) (1.3177 g, 6.24 mmol, 1.2 equiv) in 5 mL of THF were added dropwise to the solution over 2 min at 0 °C with stirring and then rinsed with 5 mL of THF. After stirred at 0 °C for 2 min, the resulting mixture was allowed to warm up to room temperature naturally and stirred for 17.5 h at rt as monitored by TLC. The resulting mixture was concentrated *in vacuo* and the residue was purified *via* chromatography on silica gel to afford (R_a,2*E*,4*E*)-**3a** (1.3041 g, 70%) (eluent: petroleum ether/ethyl acetate = 30/1) as a liquid: 97% *ee* (HPLC conditions: chiralcel OJ-H column, hexane/*i*-PrOH = 98/2, 0.7 mL/min, λ = 214 nm, t_R(minor) = 18.98 min, t_R(major) = 20.36 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.70 (d, J = 8.0 Hz, 2H, ArH), 7.29 (d, J = 8.0 Hz, 2H, ArH), 6.06 (dd, J_1 = 14.6 Hz, J_2 = 10.6 Hz, 1H, HC=), 6.01—5.89 (m, 1H, HC=), 5.71—5.58 (m, 1H, HC=), 5.37—5.28 (m, 1H, HC=), 5.13—5.04 (m, 1H, HC=), 4.87—4.79 (m, 1H, HC=), 3.85 (d, J = 7.2 Hz, 2H, NCH_2), 3.80 (dd, J_1 = 7.0 Hz, J_2 = 1.8 Hz, 2H, NCH_2), 2.42 (s, 3H, CH_3), 2.00—1.90 (m, 2H, CH_2), 1.74 (d, J = 6.4 Hz, 3H, CH_3), 1.40—1.26 (m, 4H, 2 × CH_2), 0.88 (t, J = 7.0 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 205.1, 143.0, 137.5, 134.5, 130.4, 130.1, 129.5, 127.05, 126.99, 124.2, 92.3, 86.1, 48.2, 46.0, 31.2, 28.2, 22.0, 21.4, 18.0, 13.8; MS (ESI) *m*/*z*: 382 ([M+Na]$^+$); IR (neat) *v*: 3022, 2957, 2927, 2857, 1962, 1660, 1598, 1494, 1441, 1342, 1305, 1156, 1092 cm^{-1}; HRMS calcd for $C_{21}H_{30}NO_2S$ ([M+H]$^+$): 360.1992, found: 360.1991.

(2) Preparation of (R_a)-*N*-(5-phenyl-2,3-pentadien-1-yl)-*N*-[(2*E*,4*E*)-hexadien-1-yl] toluenesulfonamide (R_a,2*E*,4*E*)-3b (hanyl-9-078)



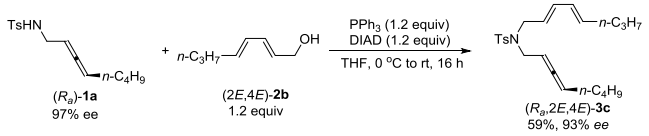
Typical procedure B: To a dry round-bottom flask placed in an ice bath were added PPh_3 (0.5711 g, 2.16 mmol, 1.2 equiv), (R_a)-*N*-(5-phenyl-2,3-pentadien-1-yl)toluenesulfonamide (R_a)-**1b** (92% *ee*, 0.5890 g, 1.8 mmol)/THF (5 mL), (2*E*,4*E*)-hexa-2,4-dien-1-ol **2a** (0.2200 g, 2.16 mmol, 1.2 equiv)/THF (3 mL), and DIAD (0.4456 g, 2.16 mmol, 1.2 equiv)/THF (2 mL) sequentially at 0 °C. The resulting mixture was allowed to warm up to room temperature naturally and stirred for 20 h at rt as monitored by TLC. Then the resulting mixture was concentrated *in vacuo* and purified *via* chromatography on silica gel to afford (R_a,2*E*,4*E*)-**3b** (0.3280 g, 44%) (eluent: petroleum ether/ethyl acetate =20/1) as a liquid: 97%

 © 2019 SIOC, CAS, Shanghai, & WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

16147065, 2019, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.201900023 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

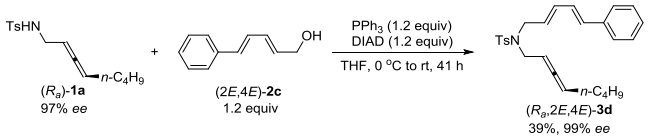
ee (HPLC conditions: chiralcel AD-H column, hexane/*i*-PrOH = 95/5, 1.0 mL/min, λ = 214 nm, t_R(minor) = 18.25 min, t_R(major) = 19.92 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.68 (d, J = 8.0 Hz, 2H, ArH), 7.26 (d, J = 6.4 Hz, 4H, ArH), 7.22—7.14 (m, 3H, ArH), 6.07-5.87 (m, 2H, 2 × HC=), 5.67—5.52 (m, 1H, HC=), 5.35—5.23 (m, 2H, 2 × HC=), 4.96—4.87 (m, 1H, HC=), 3.88—3.71 (m, 4H, 2 × NCH_2), 3.30 (dd, J_1 = 7.2 Hz, J_2 = 1.6 Hz, 2H, CH_2), 2.41 (s, 3H, CH_3), 1.71 (d, J = 6.4 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 205.6, 143.1, 139.6, 137.6, 134.6, 130.35, 130.31, 129.6, 128.37, 128.35, 127.1, 126.3, 124.1, 91.8, 86.9, 48.2, 45.8, 35.1, 21.4, 18.0; MS (ESI) *m*/*z*: 416 ([M+Na]$^+$); IR (neat) *v*: 3023, 2916, 2855, 1962, 1658, 1598, 1493, 1443, 1337, 1154, 1091 cm^{-1}; HRMS calcd for $C_{24}H_{28}O_2NS$ ([M+H]$^+$): 394.1835, found: 394.1837.

(3) Preparation of (R_a)-*N*-(2,3-octadien-1-yl)-*N*-[(2*E*,4*E*)-octadien-1-yl] toluenesulfonamide (R_a,2*E*,4*E*)-3c (hanyl-9-077)



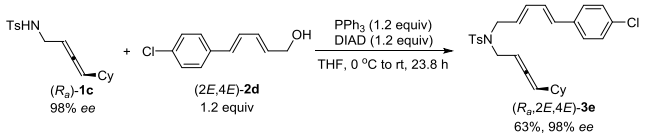
Following **Typical procedure B**, the reaction of PPh_3 (0.6334 g, 2.4 mmol, 1.2 equiv), (R_a)-*N*-(2,3-octadien-1-yl)toluenesulfonamide (R_a)-**1a** (0.5578 g, 2.0 mmol)/THF (5 mL), (2*E*,4*E*)-octa-2,4-dien-1-ol **2b** (0.3048 g, 2.4 mmol, 1.2 equiv)/THF (3 mL), DIAD (0.5063 g, 2.4 mmol, 1.2 equiv)/THF (2 mL) at rt for 16 h afforded (R_a,2*E*,4*E*)-**3c** (0.4557 g, 59%) (eluent: petroleum ether/ethyl acetate = 20/1) as a liquid: 93% *ee* (HPLC conditions: chiralcel OJ-H column, hexane/*i*-PrOH = 100/1, 1.0 mL/min, λ = 214 nm, t_R(minor) = 8.21 min, t_R(major) = 9.32 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.70 (d, J = 8.0 Hz, 2H, ArH), 7.28 (d, J = 8.4 Hz, 2H, ArH), 6.06 (dd, J_1 = 15.0 Hz, J_2 = 10.6 Hz, 1H, HC=), 5.95 (dd, J_1 = 15.2 Hz, J_2 = 10.4 Hz, 1H, HC=), 5.63 (dt, J_1 = 15.2 Hz, J_2 = 6.8 Hz, 1H, HC=), 5.34 (dt, J_1 = 15.2 Hz, J_2 = 6.8 Hz, 1H, HC=), 5.15—5.05 (m, 1H, HC=), 4.88—4.80 (m, 1H, HC=), 3.86 (d, J = 6.8 Hz, 2H, NCH_2), 3.81 (dd, J_1 = 7.0 Hz, J_2 = 2.2 Hz, 2H, NCH_2), 2.42 (s, 3H, CH_3), 2.04 (q, J = 7.1 Hz, 2H, CH_2), 2.00—1.90 (m, 2H, CH_2), 1.45—1.27 (m, 6H, 3 × CH_2), 0.92—0.85 (m, 6H, 2 × CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 205.2, 143.0, 137.6, 135.6, 134.6, 129.6, 129.2, 127.1, 124.4, 92.4, 86.1, 48.2, 46.1, 34.6, 31.2, 28.2, 22.3, 22.1, 21.4, 13.8, 13.6; MS (ESI) *m*/*z*: 410 ([M+Na]$^+$); IR (neat) *v*: 3021, 2957, 2924, 2864, 1962, 1657, 1599, 1450, 1340, 1156, 1094 cm^{-1}; HRMS calcd for $C_{23}H_{34}O_2NS$ ([M+H]$^+$): 388.2305, found: 388.2304.

(4) Preparation of (R_a)-*N*-(2,3-octadien-1-yl)-*N*-[(2*E*,4*E*)-5-phenyl-pentadien-1-yl] toluenesulfonamide (R_a,2*E*,4*E*)-3d (hanyl-9-066)



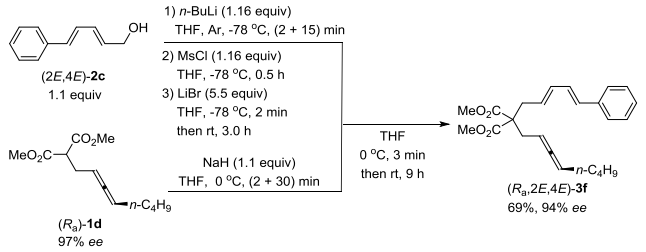
Following **Typical procedure B**, the reaction of PPh_3 (0.6369 g, 2.4 mmol, 1.2 equiv), (R_a)-*N*-(2,3-octadien-1-yl) toluenesulfonamide (R_a)-**1a** (97% *ee*, 0.5586 g, 2.0 mmol)/THF (5 mL), (2*E*,4*E*)-5-phenylpenta-2,4-dien-1-ol **2c** (0.3834 g, 2.4 mmol, 1.2 equiv)/THF (2 mL), DIAD (0.5006 g, 2.4 mmol, 1.2 equiv)/THF (3 mL) at rt for 41 h afforded (R_a,2*E*,4*E*)-**3d** (0.3316 g, 39%) (eluent: petroleum ether/ethyl acetate = 30/1) as a liquid: 98% *ee* (HPLC conditions: chiralcel AD-H column, supercritical CO_2 (l)/*i*-PrOH = 85/15, 1.3 mL/min, λ = 214 nm, t_R(minor) = 19.44 min, t_R(major) = 20.52 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.72 (d, J = 8.0 Hz, 2H, ArH), 7.36 (d, J = 7.6 Hz, 2H, Ar-H), 7.30 (t, J = 7.6 Hz, 4H, Ar-H), 7.22 (t, J = 7.2 Hz, 1H, ArH), 6.69 (dd, J_1 = 15.4 Hz, J_2 = 10.6 Hz, 1H, HC=), 6.48 (d, J = 16.0 Hz, 1H, HC=), 6.26 (dd, J_1 = 15.2 Hz, J_2 = 10.4 Hz, 1H, HC=), 5.60 (dt, J_1 = 15.2 Hz, J_2 = 6.6 Hz, 1H, HC=), 5.16—5.07 (m, 1H, HC=), 4.93—4.82 (m, 1H, HC=), 3.94 (d, J = 6.8 Hz, 2H, NCH_2), 3.84 (dd, J_1 = 6.8 Hz, J_2 = 2.0 Hz, 2H, NCH_2), 2.42 (s, 3H, CH_3), 2.03—1.90 (m, 2H, CH_2), 1.40—1.24 (m, 4H, 2 × CH_2), 0.87 (t, J = 6.8 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 205.2, 143.1, 137.5, 136.9, 134.3, 132.9, 129.6, 128.6, 127.7, 127.66, 127.1, 126.3, 92.5, 86.2, 48.3, 46.4, 31.2, 28.2, 22.1, 21.5, 13.8; MS (ESI) *m*/*z*: 865 ([2M+Na]$^+$), 444 ([M+Na]$^+$); IR (neat) *v*: 3026, 2924, 2860, 1961, 1598, 1493, 1447, 1338, 1154, 1093 cm^{-1}; HRMS calcd for $C_{26}H_{32}O_2NS$ ([M+H]$^+$): 422.2148, found: 422.2149.

(5) Preparation of (R_a)-*N*-(4-cyclohexyl-2,3-butadien-1-yl)-*N*-[(2*E*,4*E*)-5-(4-chlorophenyl)-pentadien-1-yl] toluenesulfonamide (R_a,2*E*,4*E*)-3e (hanyl-8-179)



Following **Typical procedure B**, the reaction of PPh_3 (0.3902 g, 1.44 mmol, 1.2 equiv), (R_a)-*N*-(4-cyclohexyl-2,3-butadien-1-yl) toluenesulfonamide (R_a)-**1c** (98% *ee*, 0.3663 g, 1.2 mmol)/THF (5 mL), (2*E*,4*E*)-5-(4-chlorophenyl)-2,4-pentadien-1-ol **2d** (0.2831 g, 1.44 mmol, 1.2 equiv)/THF (3 mL), DIAD (0.3008 g, 1.44 mmol, 1.2 equiv)/THF (2 mL) at rt for 23.8 h afforded (R_a,2*E*,4*E*)-**3e** (0.3667 g, 63%) (eluent: petroleum ether/ethyl acetate = 15/1) as a liquid: 98% *ee* (HPLC conditions: chiralcel AD-H column, hexane/*i*-PrOH = 90/10, 0.8 mL/min, λ = 214 nm, t_R(minor) = 10.86 min, t_R(major) = 12.03 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.71 (d, J = 8.0 Hz, 2H, ArH), 7.36—7.24 (m, 6H, ArH), 6.66 (dd, J_1 = 15.6 Hz, J_2 = 10.4 Hz, 1H, HC=), 6.43 (d, J = 16.0 Hz, 1H, HC=), 6.26 (dd, J_1 = 15.2 Hz, J_2 = 10.4 Hz, 1H, HC=), 5.63 (dt, J_1 = 15.2 Hz, J_2 = 6.6 Hz, 1H, HC=), 5.15—5.07 (m, 1H, HC=), 4.90 (qd, J_1 = 6.7 Hz, J_2 = 2.8 Hz, 1H, HC=), 3.94 (d, J = 6.8 Hz, 2H, NCH_2), 3.83 (dd, J_1 = 7.0 Hz, J_2 = 2.2 Hz, 2H, NCH_2), 2.43 (s, 3H, CH_3), 2.00—1.87 (m, 1H, CH), 1.74—1.58 (m, 5 H, 2 × CH_2 + CH), 1.32—0.98 (m, 5H, 2 × CH_2 + CH); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 204.1, 143.2, 137.5, 135.5, 133.9, 133.2, 131.5, 129.7, 128.8, 128.6, 128.4, 127.5, 127.2, 98.6, 87.2, 48.3, 46.6, 37.1, 33.0, 32.97, 25.99, 25.91, 25.88, 21.5; MS (ESI) *m*/*z*: 506 ([M(^{37}Cl)+Na]$^+$), 504 ([M(^{35}Cl)+Na]$^+$); IR (neat) v: 3029, 2923, 2850, 1959, 1596, 1490, 1446, 1405, 1341, 1305, 1156, 1090 cm^{-1}; HRMS calcd for $C_{28}H_{33}O_2N^{35}ClS$ ([M(^{35}Cl)+H]$^+$): 482.1915, found: 482.1912.

(6) Preparation of (R_a)-dimethyl 2-(2,3-octadien-1-yl)-2-[(2*E*,4*E*)-5-phenylhexadien-1-yl]malonate (R_a,2*E*,4*E*)-3f (hanyl-9-141, hanyl-9-142)



Typical procedure C: To a flame-dried Schlenk tube were added (2*E*,4*E*)-5-phenyl-penta-2,4-dien-1-ol **2c** (0.3533 g, 2.2 mmol, 1.1 equiv) and anhydrous THF (5 mL) under Ar. The Schlenk tube was then placed into a dry ice/ethanol bath (–78 °C). *n*-BuLi (0.93 mL, 2.5 mol·L^{-1} in hexane, 2.31 mmol, 1.16 equiv) was added dropwise over 2 min and the resulting solution was stirred for 15 minutes at this temperature. MsCl (0.18 mL, d = 1.48 g/mL, 0.2664 g, 2.31 mmol, 1.16 equiv) was then added and the resulting mixture was stirred for 30 min followed by the addition of LiBr (0.9812 g, 11.0 mmol, 5.5 equiv) at –78 °C and stirred at this temperature for 2 min. The resulting mixture was removed from the dry ice/ethanol bath and then allowed to warm up to room

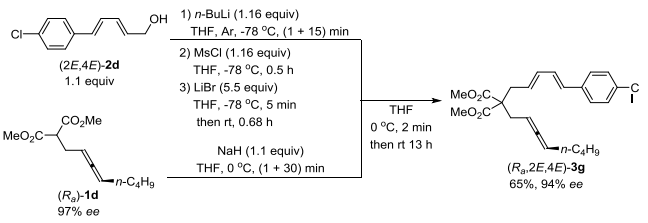
 © 2019 SIOC, CAS, Shanghai, & WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

16147065, 2019, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.201900023 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

temperature and stirred at rt for 3 h, and the solution was used directly in the next step.

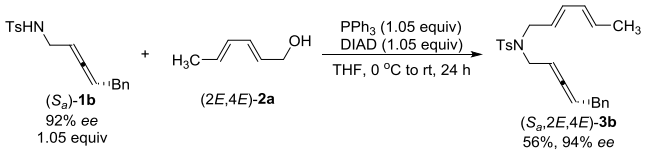
To a separate flame-dried Schlenk tube was added NaH (60% dispersion in mineral oil, 0.0884 g, 2.2 mmol, 1.1 equiv). The tube was placed into an ice bath (0 °C). To the Schlenk tube was added THF (3 mL) followed by the dropwise addition of (R_a)-dimethyl 2-(2,3-octadien-1-yl) malonate (R_a)-**1d** (97% *ee*, 0.4446 g, 2.0 mmol) over 2 min at 0 °C. Then 2 mL of THF was added and the resulting mixture was then stirred for 30 min at this temperature. The solution containing the (2*E*,4*E*)-dienyl bromide prepared above was then transferred *via* a syringe to the Schlenk tube containing the malonate and the resulting mixture was allowed to warm up to room temperature naturally and stirred for 9 h as monitored by TLC. A saturated aqueous solution of NH_4Cl (5 mL) was added to the resulting mixture to quench the reaction and white solid was precipitated out. Water (5 mL) was added to dissolve the solid and the resulting mixture was poured into a separatory funnel and then extracted with Et_2O (20 mL × 3). The organic layer was combined, washed with brine (20 mL), dried with anhydrous $MgSO_4$, filtered, and concentrated *in vacuo*. The residue was purified *via* flash column chromatography on silica gel to afford (R_a,2*E*,4*E*)-**3f** (0.4888 g, 69%) (eluent: petroleum ether/ethyl acetate = 30/1) as a liquid: 94% *ee* (HPLC conditions: chiralcel AD-H column, hexane/*i*-PrOH = 95/5, 1.0 mL/min, λ = 214 nm, t_R(major) = 5.32 min, t_R(minor) = 5.74 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.39—7.33 (m, 2H, ArH), 7.29 (t, J = 7.6 Hz, 2H, ArH), 7.20 (tt, J_1 = 7.2 Hz, J_2 = 1.5 Hz, 1H, ArH), 6.72 (dd, J_1 = 15.8 Hz, J_2 = 10.6 Hz, 1H, HC=), 6.45 (d, J = 15.6 Hz, 1H, HC=), 6.25 (dd, J_1 = 15.0 Hz, J_2 = 10.6 Hz, 1H, HC=), 5.63 (dt, J_1 = 15.2 Hz, J_2 = 7.6 Hz, 1H, HC=), 5.09 (qt, J_1 = 6.7 Hz, J_2 = 2.2 Hz, 1H, HC=), 4.96—4.86 (m, 1H, HC=), 3.72 (s, 6H, 2 × CH_3), 2.78 (d, J = 8.0 Hz, 2H, CH_2), 2.61 (dd, J_1 = 7.6 Hz, J_2 = 2.4 Hz, 2H, CH_2), 1.98 (qd, J_1 = 7.0 Hz, J_2 = 2.8 Hz, 2H, CH_2), 1.44—1.27 (m, 4H, 2 × CH_2), 0.89 (t, J = 7.2 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 205.8, 171.0, 137.1, 134.6, 131.5, 128.54, 128.50, 127.9, 127.4, 126.2, 91.0, 84.5, 58.0, 52.4, 52.36, 35.9, 32.7, 31.3, 28.4, 22.1, 13.8; MS (ESI) *m/z*: 787 ([2M+Na]$^+$), 405 ([M+Na]$^+$), 383 ([M+H]$^+$); IR (neat) *v*: 3023, 2952, 2860, 1962, 1733, 1594, 1440, 1279, 1190, 1075 cm^{-1}; HRMS calcd for $C_{24}H_{31}O_4$ ([M+H]$^+$): 383.2217, found: 383.2215.

(7) Preparation of (R_a)-dimethyl 2-(2,3-octadien-1-yl)-2-((2*E*,4*E*)-5-(4-chlorophenyl)pentadien-1-yl)malonate (R_a,2*E*,4*E*)-3g (hanyl-9-131, hanyl-9-132)



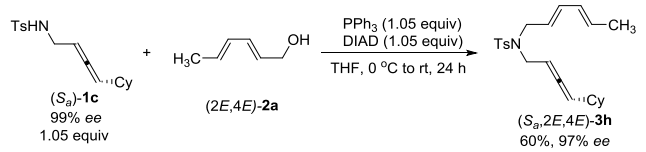
Following **Typical procedure C**, the reaction of (2*E*,4*E*)-5-(4-chlorophenyl)-2,4-pentadien-1-ol **2d** (0.4321 g, 2.2 mmol, 1.16 equiv), *n*-BuLi (0.93 mL, 2.5 mol·L^{-1} in hexane, 2.31 mmol, 1.16 equiv), MsCl (0.18 mL, 0.2664 g, 2.31 mmol, 1.16 equiv), LiBr (0.9676 g, 11.0 mmol, 5.5 equiv), NaH (0.0912 g, 2.2 mmol, 1.1 equiv), and (R_a)-dimethyl 2-(2,3-octadien-1-yl)malonate (R_a)-**1d** (97% *ee*, 0.4803 g, 2.0 mmol) in THF (10 mL) afforded (R_a,2*E*,4*E*)-**3g** (0.5397 g, 65%) (eluent: petroleum ether/ethyl acetate = 30/1) as an oil: 94% *ee* (HPLC conditions: chiralcel AD-H column, hexane/*i*-PrOH = 95/5, 0.7 mL/min, λ = 214 nm, t_R(major) = 9.69 min, t_R(minor) = 10.72 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.32—7.23 (m, 4H, ArH), 6.69 (dd, J_1 = 15.6 Hz, J_2 = 10.4 Hz, 1H, HC=), 6.40 (d, J = 15.6 Hz, 1H, HC=), 6.23 (dd, J_1 = 14.8 Hz, J_2 = 10.4 Hz, 1H, HC=), 5.65 (dt, J_1 = 15.2 Hz, J_2 = 7.6 Hz, 1H, CH), 5.15—5.05 (m, 1H, HC=), 4.95—4.85 (m, 1H, HC=), 3.73 (s, 6H, 2 × CH_3), 2.78 (d, J = 7.6 Hz, 2H, CH_2), 2.61 (dd, J_1 = 7.8 Hz, J_2 = 2.2 Hz, 2H, CH_2), 2.03—1.93 (m, 2H, CH_2), 1.45—1.28 (m, 4H, 2 × CH_2), 0.89 (t, J = 7.0 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 205.8, 171.0, 135.7, 134.3, 132.9, 130.2, 129.2, 128.69, 128.68, 127.3, 91.1, 84.5, 58.0, 52.46, 52.43, 35.9, 32.7, 31.3, 28.5, 22.1, 13.9; MS (ESI) *m/z*: 441 ([M(^{37}Cl)+Na]$^+$), 439 ([M(^{35}Cl)+Na]$^+$), 419 ([M(^{37}Cl)+H]$^+$), 417 ([M(^{35}Cl)+H]$^+$); IR (neat) *v*: 3023, 2952, 2859, 1962, 1733, 1488, 1439, 1278, 1189, 1084 cm^{-1}; HRMS calcd for $C_{24}H_{30}{}^{35}ClO_4$ ([M(^{35}Cl)+H]$^+$): 417.1827, found: 417.1825.

(8) Preparation of (S_a)-*N*-(5-phenyl-2,3-pentadien-1-yl)-*N*-[(2*E*,4*E*)-hexadien-1-yl] toluenesulfonamide (S_a,2*E*,4*E*)-3b (hanyl-10-008)



Following **Typical procedure B**, the reaction of PPh_3 (0.5577 g, 2.1 mmol, 1.05 equiv), (S_a)-*N*-(5-phenyl-2,3-pentadien-1-yl) toluenesulfonamide (S_a)-**1b** (92% *ee*, 0.6587 g, 2.1 mmol, 1.05 equiv)/THF (5 mL), (2*E*,4*E*)-hexa-2,4-dien-1-ol **2a** (0.1986 g, 2.0 mmol)/THF (3 mL), DIAD (0.4575 g, 2.1 mmol, 1.05 equiv)/THF (2 mL) at rt for 24 h afforded (S_a,2*E*,4*E*)-**3g** (0.4484 g, 56%) (eluent: petroleum ether/ethyl acetate = 20/1) as a liquid: 94% *ee* (HPLC conditions: chiralcel AS-H column, hexane/*i*-PrOH = 95/5, 1.0 mL/min, λ = 214 nm, t_R(minor) = 31.02 min, t_R(major) = 37.67 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.68 (d, J = 8.4 Hz, 2H, ArH), 7.32—7.25 (m, 4H, ArH), 7.23—7.14 (m, 3H, ArH), 6.07—5.89 (m, 2H, 2 × HC=), 5.64—5.53 (m, 1H, HC=), 5.33—5.24 (m, 2H, 2 × HC=), 4.95—4.87 (m, 1H, HC=), 3.88—3.71 (m, 4H, 2 × NCH_2), 3.30 (dd, J_1 = 7.4 Hz, J_2 = 2.6 Hz, 2H, CH_2), 2.42 (s, 3H, CH_3), 1.71 (d, J = 6.8 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 205.6, 143.1, 139.7, 137.7, 134.6, 130.4, 129.7, 128.45, 128.42, 127.2, 126.3, 124.2, 91.9, 87.0, 48.3, 45.8, 35.2, 21.5, 18.1; MS (ESI) *m/z*: 416 ([M+Na]$^+$); IR (neat) *v*: 3665, 3557, 3024, 2914, 2853, 2733, 2584, 1963, 1809, 1660, 1599, 1494, 1453, 1340, 1305, 1156, 1092, 1029 cm^{-1}; HRMS calcd for $C_{24}H_{28}NO_2S$ ([M+H]$^+$): 394.1835, found: 394.1832.

(9) Preparation of (S_a)-*N*-(4-cyclohexyl-2,3-butadien-1-yl)-*N*-[(2*E*,4*E*)-hexadien-1-yl] toluenesulfonamide (S_a,2*E*,4*E*)-3h (hanyl-9-200)



Following **Typical procedure B**, the reaction of PPh_3 (0.5518 g, 2.1 mmol, 1.05 equiv), (S_a)-*N*-(4-cyclohexyl-2,3-butadien-1-yl)-toluenesulfonamide (S_a)-**1c** (99% *ee*, 0.6420 g, 2.1 mmol, 1.05 equiv)/THF (5 mL), (2*E*,4*E*)-hexa-2,4-dien-1-ol **2a** (0.1970 g, 2.0 mmol)/THF (3 mL), DIAD (0.4366 g, 2.1 mmol, 1.05 equiv)/THF (2 mL) at rt for 24 h afforded (S_a,2*E*,4*E*)-**3c** (0.4463 g, 60%) (eluent: petroleum ether/ethyl acetate =30/1) as a liquid: 97% *ee* (HPLC conditions: chiralcel AD-H column, hexane/*i*-PrOH = 98/2, 1.0 mL/min, λ = 214 nm, t_R(major) = 15.27 min, t_R(minor) = 16.76 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.70 (d, J = 8.4 Hz, 2H, ArH), 7.28 (d, J = 8.0 Hz, 2H, ArH), 6.06 (dd, J_1 = 14.8 Hz, J_2 = 10.4 Hz, 1H, HC=), 6.01—5.92 (m, 1H, HC=), 5.70—5.58 (m, 1H, HC=), 5.32 (dt, J_1 = 14.8 Hz, J_2 = 6.8 Hz, 1H, HC=), 5.11—5.06 (m, 1H, HC=), 4.88 (qd, J_1 = 6.7 Hz, J_2 = 2.8 Hz, 1H, HC=), 3.86 (d, J = 6.8 Hz, 2H, NCH_2), 3.80 (dd, J_1 = 7.0 Hz, J_2 = 2.2 Hz, 2H, NCH_2), 2.42 (s, 3H, CH_3), 1.99—1.87 (m, 1H, CH of Cy), 1.76—1.58 (m, 8H, CH_3 + 2 × CH_2 + one proton of CH_2), 1.31—0.96 (m, 5H, 2 × CH_2 + one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 204.0, 143.0, 137.6, 134.5, 130.4, 130.2, 129.6, 127.1, 124.3, 98.4, 87.2, 48.2, 46.2, 37.0, 32.9, 26.0, 25.9, 21.5, 18.0; MS (ESI) *m/z*: 408 ([M+Na]$^+$); IR (neat) *v*:

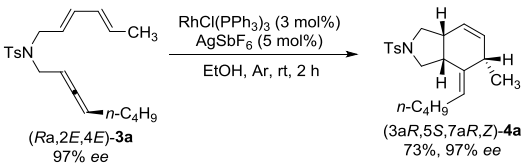
 © 2019 SIOC, CAS, Shanghai, & WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

16147065, 2019, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.201900023 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

3666, 3553, 3370, 3021, 2923, 2851, 2733, 2659, 2589, 2483, 2364, 2304, 2247, 1960, 1918, 1811, 1729, 1661, 1598, 1494, 1447, 1344, 1305, 1261, 1225, 1157, 1119, 1093, 1047, 1019 cm^{-1}; HRMS calcd for $C_{23}H_{32}NO_2S$ ($[M+H]^+$): 386.2148, found: 386.2143.

Rhodium-catalyzed highly diastereoselective intramolecular [4+2] cycloaddition of allene-1,3-dienes

(1) Preparation of (3a*R*,5*S*,7a*R,Z*)-5-methyl-2-((4-methylphenyl)-sulfonyl)-4-pentylidene-2,3,3a,4,5,7a-hexahydro-1*H*-isoindole (3a*R*,5*S*,7a*R,Z*)-4a (hanyl-12-162)

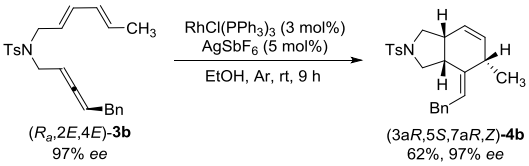


(*R*a,2*E*,4*E*)-**3a** 97% *ee*

(3a*R*,5*S*,7a*R*,*Z*)-**4a** 73%, 97% *ee*

Typical procedure I: To a flame-dried Schlenk tube were added $RhCl(PPh_3)_3$ (14.2 mg, 0.015 mmol) and $AgSbF_6$ (8.9 mg, 0.025 mmol) inside a glove box. Under argon atmosphere, to the Schlenk tube was added 3 mL of EtOH outside the glove box. Then the resulting mixture was stirred at room temperature for 10 min. To the tube were added (R_a,2*E*,4*E*)-**3a** (178.6 mg, 0.5 mmol, 97% *ee*) and EtOH (2 mL) sequentially. After being stirred at room temperature for 2 h, the reaction was complete as monitored by TLC. The crude reaction mixture was filtrated through a pad of silica gel and eluted with diethyl ether (20 mL) and then concentrated *in vacuo* to afford the crude product, which was separated by chromatography on silica gel to afford (3a*R*,5*S*,7a*R,Z*)-**4a** (131.1 mg, 73%) (eluent: petroleum ether/ethyl acetate = 30/1) as s liquid: 97% *ee* (HPLC conditions: chiralcel OD-H column, hexane/*i*-PrOH = 98/2, 1.5 mL/min, λ = 214 nm, t_R(major) = 8.50 min, t_R(minor) = 11.87 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.71 (d, J = 8.0 Hz, 2H, ArH), 7.32 (d, J = 8.0 Hz, 2H, ArH), 5.44—5.34 (m, 2H, 2 × HC=), 5.12 (d, J = 10.0 Hz, 1H, HC=), 3.49 (dd, J_1 = 10.0 Hz, J_2 = 6.8 Hz, 1H, one proton of NCH_2), 3.36 (t, J = 8.4 Hz, 1H, one proton of NCH_2), 3.29 (d, J = 10.0 Hz, 1H, one proton of NCH_2), 3.21—3.13 (m, 1H, CH), 3.11—3.04 (m, 1H, one proton of NCH_2), 2.79—2.67 (m, 1H, CH), 2.64—2.57 (m, 1H, CH), 2.44 (s, 3H, CH_3), 2.02—1.91 (m, 2H, CH_2), 1.35—1.22 (m, 4H, 2 × CH_2), 1.07 (d, J = 7.6 Hz, 3H, CH_3), 0.88 (t, J = 6.8 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 143.3, 135.6, 133.9, 133.1, 129.4, 129.3, 127.5, 125.8, 54.5, 50.6, 39.2, 38.1, 35.2, 32.1, 26.7, 24.9, 22.2, 21.5, 13.9; MS (EI) *m/z* (%): 359 (M^+, 1.02), 91 (100); IR (neat) *v*: 2956, 2924, 2894, 2859, 1451, 1335, 1300, 1290, 1232, 1161, 1131, 1087, 1052, 1028, 1012 cm^{-1}; HRMS calcd for $C_{21}H_{29}NO_2S$ (M^+): 359.1914, found: 359.1915.

(2) Preparation of (3a*R*,5*S*,7a*R,Z*)-5-methyl-4-phenylethylidene-2-((4-methylphenyl)sulfonyl)-2,3,3a,4,5,7a-hexahydro-1*H*-isoindole (3a*R*,5*S*,7a*R,Z*)-4b (hanyl-9-109)

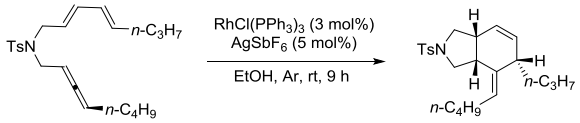


(R_a,2*E*,4*E*)-**3b** 97% *ee*

(3a*R*,5*S*,7a*R*,*Z*)-**4b** 62%, 97% *ee*

Following **Typical procedure I**, the reaction of $RhCl(PPh_3)_3$ (13.6 mg, 0.015 mmol), $AgSbF_6$ (8.7 mg, 0.025 mmol), and (R_a,2*E*,4*E*)-**3b** (196.7 mg, 0.5 mmol, 97% *ee*) in EtOH (5.0 mL) afforded (3a*R*,5*S*,7a*R,Z*)-**4b** (121.6 mg, 62%) (eluent: petroleum ether/ethyl acetate = 15/1) as a white solid: m.p. 117—120 °C (dichloromethane/petroleum ether); 97% *ee* (HPLC conditions: chiralcel OD-H column, hexane/*i*-PrOH = 95/5, 1.0 mL/min, λ = 214 nm, t_R(major) = 16.45 min, t_R(minor) = 22.78 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.70 (d, J = 8.4 Hz, 2H, ArH), 7.29 (t, J = 8.4 Hz, 4H, ArH), 7.20 (t, J = 7.2 Hz, 1H, ArH), 7.10 (d, J = 7.2 Hz, 2H, ArH), 5.59 (t, J = 7.6 Hz, 1H, HC=), 5.43 (dt, J_1 = 10.0 Hz, J_2 = 2.9 Hz, 1H, HC=), 5.16 (d, J = 10.0 Hz, 1H, HC=), 3.48 (dd, J_1 = 10.0 Hz, J_2 = 6.4 Hz, 1H, one proton of NCH_2), 3.41—3.21 (m, 5H, CH_2 of Bn + two protons of NCH_2 + CH), 3.14 (dd, J_1 = 10.4 Hz, J_2 = 9.2 Hz, 1H, one proton of NCH_2), 2.86—2.76 (m, 1H, CH), 2.70—2.62 (m, 1H, CH), 2.42 (s, 3H, CH_3), 1.12 (d, J = 7.6 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 143.3, 140.6, 137.2, 133.8, 132.9, 129.5, 128.5, 128.1, 127.5, 127.2, 126.0,125.8, 54.5, 50.5, 39.3, 38.1, 35.3, 33.1, 24.8, 21.5; MS (ESI) *m/z*: 416 ($[M+Na]^+$), 394 ($[M+H]^+$); IR (neat) *v*: 3078, 3015, 2955, 2908, 2865, 1596, 1490, 1451, 1381, 1337, 1292, 1229, 1177, 1153, 1130, 1105, 1086, 1052, 1023 cm^{-1}; Anal Calcd for $C_{24}H_{27}NO_2S$: C 73.25, H 6.92, N 3.56. found: C 73.43, H 7.26, N 3.26.

(3) Preparation of (3a*R*,5*S*,7a*R,Z*)-4-phenylethylidene-5-propyl-2-((4-methylphenyl)sulfonyl)-2,3,3a,4,5,7a-hexahydro-1*H*-isoindole (3a*R*,5*S*,7a*R,Z*)-4c (hanyl-9-111)

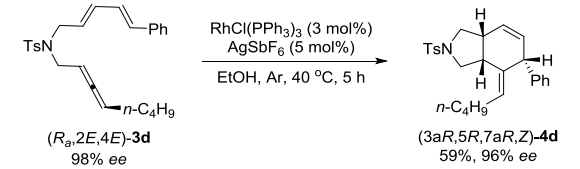


(R_a,2*E*,4*E*)-**3c** 93% *ee*

(3a*R*,5*S*,7a*R*,*Z*)-**4c** 75%, 93% *ee*

Following **Typical procedure I**, the reaction of $RhCl(PPh_3)_3$ (11.0 mg, 0.012 mmol), $AgSbF_6$ (6.9 mg, 0.02 mmol), and (R_a,2*E*,4*E*)-**3c** (157.5 mg, 0.4 mmol, 93% *ee*) in EtOH (4.0 mL) afforded (3a*R*,5*S*,7a*R,Z*)-**4c** (118.4 mg, 75%) (eluent: petroleum ether/ethyl acetate = 30/1) as a liquid: 93% *ee* (HPLC conditions: chiralcel OD-H column, hexane/*i*-PrOH = 100/1, 1.0 mL/min, λ = 214 nm, t_R(major) = 15.71 min, t_R(minor) = 19.35 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.70 (d, J = 8.0 Hz, 2H, ArH), 7.31 (d, J = 8.0 Hz, 2H, ArH), 5.54—5.48 (m, 1H, HC=), 5.32 (t, J = 7.4 Hz, 1H, HC=), 5.16 (d, J = 10.0 Hz, 1H, HC=), 3.49 (dd, J_1 = 10.4 Hz, J_2 = 6.8 Hz, 1H, one proton of NCH_2), 3.34 (t, J = 8.8 Hz, 1H, one proton of NCH_2), 3.24 (dd, J_1 = 10.2 Hz, J_2 = 1.8 Hz, 1H, one proton of NCH_2), 3.21—3.14 (m, 1H, CH), 3.04 (t, J = 10.0 Hz, 1H, one proton of NCH_2), 2.68—2.61 (m, 1H, CH), 2.58—2.50 (m, 1H, CH), 2.44 (s, 3H, CH_3), 1.97 (q, J = 6.9 Hz, 2H, CH_2), 1.36—1.18 (m, 8H, 4 × CH_2), 0.92—0.84 (m, 6H, 2 × CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 143.3, 134.2, 133.6, 131.7, 129.6, 129.4, 127.6, 126.0, 54.5, 50.5, 41.0, 40.8, 39.6, 37.8, 32.1, 26.8, 22.2, 21.5, 20.9, 14.0, 13.9; MS (ESI) *m/z*: 797 ($[2M+Na]^+$), 410 ($[M+Na]^+$), 388 ($[M+H]^+$); IR (neat) *v*: 3015, 2953, 2923, 2861, 1654, 1597, 1460, 1385, 1333, 1222, 1157, 1120, 1094, 1030 cm^{-1}; HRMS calcd for $C_{23}H_{34}O_2NS([M+H]^+)$: 388.2305, found: 388.2305.

(4) Preparation of (3a*R*,5*R,7a*R,Z*)-4-pentylidene-5-phenyl-2-((4-methylphenyl)sulfonyl)-2,3,3a,4,5,7a-hexahydro-1*H*-isoindole (3a*R*,5*R,7a*R,Z*)-4d (hanyl-9-138)



(R_a,2*E*,4*E*)-**3d** 98% *ee*

(3a*R*,5*R*,7a*R*,*Z*)-**4d** 59%, 96% *ee*

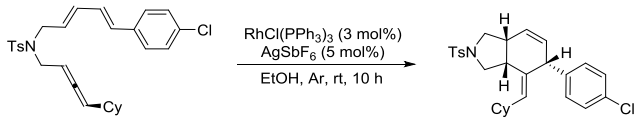
Following **Typical procedure I**, the reaction of $RhCl(PPh_3)_3$ (13.8 mg, 0.015 mmol), $AgSbF_6$ (8.6 mg, 0.025 mmol), and (R_a,2*E*,4*E*)-**3d** (184.8 mg, 0.5 mmol, 98% *ee*) in EtOH (5 mL) afforded (3a*R*,5*R*,7a*R,Z*)-**4d** (110.0 mg, 59%) (eluent: petroleum ether/ethyl acetate = 15/1) as a liquid: 96% *ee* (HPLC conditions: chiralcel OD-H column, hexane/*i*-PrOH = 95/5, 1.0 mL/min, λ = 214 nm, t_R(major) = 11.04 min, t_R(minor) = 13.57 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.56 (d, J = 8.0 Hz, 2H, ArH), 7.27 (d, J = 8.0 Hz, 2H, ArH), 7.24—7.15 (m, 3H, ArH), 7.13—7.07 (m, 2H, ArH), 5.81 (ddd, J_1 = 10.2 Hz, J_2 = 4.2 Hz, J_3 = 2.4 Hz, 1H, HC=), 5.58 (t, J = 7.4 Hz, 1H, HC=), 5.48 (dt, J_1 = 10.3 Hz, J_2 = 1.9 Hz, 1H, HC=), 3.92 (br, 1H, one

 © 2019 SIOC, CAS, Shanghai, & WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

16147065, 2019, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.201900023 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

proton of NCH_2), 3.55 (dd, J_1 = 10.2 Hz, J_2 = 7.2 Hz, 1H, one proton of NCH_2), 3.34—3.25 (m, 1H, CH), 3.20 (dd, J_1 = 10.4 Hz, J_2 = 2.4 Hz, 1H, one proton of NCH_2), 3.07 (dd, J_1 = 9.8 Hz, J_2 = 8.6 Hz, 1H, one proton of NCH_2), 2.83—2.76 (m, 1H, CH), 2.72 (t, J = 10.2 Hz, 1H, CH), 2.45 (s, 3H, CH_3), 2.08—1.98 (m, 2H, CH_2), 1.37—1.22 (m, 4H, 2 × CH_2), 0.88 (t, J = 7.2 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 143.6, 143.1, 133.7, 133.3, 130.9, 129.7, 129.4, 128.5, 128.1, 127.5, 127.0, 126.1, 53.9, 49.6, 45.6, 39.7, 38.0, 32.0, 27.0, 22.2, 21.5, 13.9; MS (ESI) *m/z*: 444 ([M+Na]$^+$), 422 ([M+H]$^+$); IR (neat) *v*: 3025, 2954, 2861, 1597, 1483, 1455, 1340, 1214, 1159, 1094, 1042 cm^{-1}; HRMS calcd for $C_{26}H_{32}NO_2S$ ([M+H]$^+$): 422.2148, found: 422.2144.

(5) Preparation of (3a*R*,5*R*,7a*R*,*Z*)-5-(4-chlorophenyl)-4-cyclohexylmethylene-2-((4-methylphenyl)sulfonyl)-2,3,3a,4,5,7a-hexahydro-1*H*-isoindole (3a*R*,5*R*,7a*R*,*Z*)-4e (hanyl-9-019)

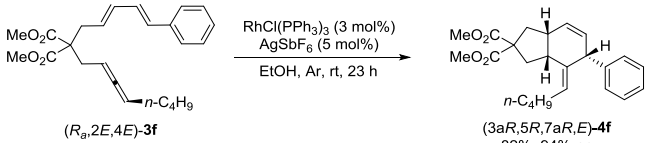


(R_a,2*E*,4*E*)-**3e** 98% *ee*

(3a*R*,5*R*,7a*R*,*Z*)-**4e** 79%, 99% *ee*

Following **Typical procedure I**, the reaction of $RhCl(PPh_3)_3$ (14.0 mg, 0.015 mmol), $AgSbF_6$ (8.6 mg, 0.025 mmol), and (R_a,2*E*,4*E*)-**3e** (247.0 mg, 0.5 mmol, 98% *ee*) in EtOH (5.0 mL) afforded (3a*R*,5*R*,7a*R*,*Z*)-**4e** (194.1 mg, 79%) (eluent: petroleum ether/ethyl acetate = 20/1) as a white solid: 99% *ee* (HPLC conditions: chiralcel AD-H column, hexane/*i*-PrOH = 90/10, 1.0 mL/min, λ = 214 nm, t_R(major) = 6.68 min, t_R(minor) = 7.78 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.55 (d, J = 8.4 Hz, 2H, ArH), 7.29 (d, J = 8.0 Hz, 2H, ArH), 7.13 (d, J = 8.8 Hz, 2H, ArH), 7.02 (d, J = 8.4 Hz, 2H, ArH), 5.80 (ddd, J_1 = 10.2 Hz, J_2 = 4.4 Hz, J_3 = 2.2 Hz, 1H, HC=), 5.54 (d, J = 10.0 Hz, 1H, HC=), 5.41 (d, J = 9.2 Hz, 1H, HC=), 3.83 (brs, 1H, one proton of NCH_2), 3.59 (dd, J_1 = 10.0 Hz, J_2 = 7.2 Hz, 1H, one proton of NCH_2), 3.32 (dd, J_1 = 18.4 Hz, J_2 = 8.4 Hz, 1H, CH), 3.11 (dd, J_1 = 10.0 Hz, J_2 = 2.8 Hz, 1H, one proton of NCH_2), 3.06 (t, J = 9.0 Hz, 1H, one proton of NCH_2), 2.87—2.79 (m, 1H, CH), 2.53 (t, J = 10.2 Hz, 1H, CH), 2.48 (s, 3H, CH_3), 2.25—2.13 (m, 1H, CH of Cy), 1.77—1.48 (m, 5H, 2 × CH_2 + one proton of CH_2), 1.34—1.02 (m, 5H, 2 × CH_2 + one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 143.3, 142.1, 137.4, 133.1, 131.8, 131.77, 129.5, 129.1, 129.0, 128.3, 128.2, 127.6, 53.9, 50.0, 45.0, 39.7, 38.2, 36.5, 33.8, 33.3, 25.8, 25.7, 21.5; MS (ESI) *m/z*: 484 ([M(^{37}Cl)+H]$^+$), 482 ([M(^{35}Cl)+H]$^+$); IR (neat) *v*: 2921, 2848, 1597, 1487, 1447, 1403, 1344, 1305, 1160, 1090, 1041, 1013 cm^{-1}; HRMS calcd for $C_{28}H_{33}{}^{35}ClNO_2S$ ([M(^{35}Cl)+H]$^+$): 482.1915, found: 482.1915.

(6) Preparation of (3a*R*,5*R*,7a*R*,*E*)-dimethyl 5-phenyl-4-pentyl-idene-1,3,3a,4,5,7a-hexahydro-2*H*-indene-2,2-dicarboxylate (3a*R*,5*R*,7a*R*,*E*)-4f (hanyl-9-147)

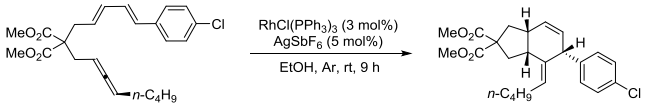


(R_a,2*E*,4*E*)-**3f** 94% *ee*

(3a*R*,5*R*,7a*R*,*E*)-**4f** 82%, 94% *ee*

Following **Typical procedure I**, the reaction of $RhCl(PPh_3)_3$ (13.8 mg, 0.015 mmol), $AgSbF_6$ (8.5 mg, 0.025 mmol), and (R_a,2*E*,4*E*)-**3f** (194.4 mg, 0.5 mmol, 94% *ee*) in EtOH (5.0 mL) was carried out at rt for 23 h (the reaction process is tracked *via* NMR analysis). Separation of the crude mixture by chromatography on silica gel afforded (3a*R*,5*R*,7a*R*,*E*)-**4f** (158.7 mg, 82%) (eluent: petroleum ether/ethyl acetate = 30/1) as a liquid: 94% *ee* (HPLC conditions: chiralcel OD-H column, hexane/*i*-PrOH = 200/1, 1.0 mL/min, λ = 214 nm, t_R(major) = 11.86 min, t_R(minor) = 19.35 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.33 (d, J = 7.6 Hz, 2H, ArH), 7.26 (t, J = 7.6 Hz, 2H, ArH), 7.16 (t, J = 7.0 Hz, 1H, ArH), 5.92 (ddd, J_1 = 10.0 Hz, J_2 = 4.4 Hz, J_3 = 2.4 Hz, 1H, HC=), 5.75 (d, J = 10.0 Hz, 1H, HC=), 5.56 (t, J = 7.4 Hz, 1H, HC=), 3.98 (brs, 1H, one proton of CH_2), 3.69 (s, 3H, OCH_3), 3.63 (s, 3H, OCH_3), 3.24 (dt, J_1 = 12.4 Hz, J_2 = 7.6 Hz, 1H, one proton of CH_2), 2.82—2.74 (m, 1H, CH), 2.54 (dd, J_1 = 13.6 Hz, J_2 = 8.4 Hz, 1H, one proton of CH_2), 2.30 (dd, J_1 = 13.8 Hz, J_2 = 3.4 Hz, 1H, one proton of CH_2), 2.15—2.00 (m, 3H, one proton of CH_2 + CH_2), 1.94 (t, J = 13.0 Hz, 1H, one proton of CH_2), 1.42—1.27 (m, 4H, 2 × CH_2), 0.90 (t, J = 7.2 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 172.9, 172.2, 144.4, 136.8, 131.0, 129.1, 128.0, 127.9, 127.2, 125.8, 59.5, 52.6, 52.4, 46.1, 40.3, 40.1, 38.9, 37.7, 32.1, 26.9, 22.2, 13.9; MS (ESI) *m/z*: 405 ([M+Na]$^+$), 383 ([M+H]$^+$); IR (neat) *v*: 3021, 2951, 2863, 1733, 1596, 1491, 1442, 1247, 1195, 1162, 1110, 1064 cm^{-1}; HRMS calcd for $C_{24}H_{34}NO_4$ ([M+NH_4]$^+$): 400.2482, found: 400.2480.

(7) Preparation of (3a*R*,5*R*,7a*R*,*E*)-dimethyl 5-phenyl-4-pentylidene-1,3,3a,4,5,7a-hexahydro-2*H*-indene-2,2-dicarboxylate (3a*R*,5*R*,7a*R*,*E*)-4g (hanyl-9-134)

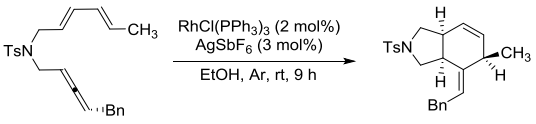


(R_a,2*E*,4*E*)-**3g** 94% ee

(3a*R*,5*R*,7a*R*,*E*)-**4g** 80%, 94% *ee*

Following **Typical procedure I**, the reaction of $RhCl(PPh_3)_3$ (13.8 mg, 0.015 mmol), $AgSbF_6$ (8.7 mg, 0.025 mmol), and (R_a,2*E*,4*E*)-**3g** (208.1 mg, 0.5 mmol, 94% *ee*) in EtOH (5.0 mL) at rt for 9 h (the reaction process is tracked *via* NMR analysis) afforded (3a*R*,5*R*,7a*R*,*E*)-**4g** (167.4 mg, 80%) (eluent: petroleum ether/ethyl acetate = 30/1) as a liquid: 94% *ee* (HPLC conditions: chiralcel OD-H column, hexane/*i*-PrOH = 95/5, 0.7 mL/min, λ = 214 nm, t_R(major) = 6.50 min, t_R(minor) = 7.89 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.30—7.20 (m, 4H, ArH), 5.88 (ddd, J_1 = 10.0 Hz, J_2 = 4.2 Hz, J_3 = 2.2 Hz, 1H, HC=), 5.76 (d, J = 10.4 Hz, 1H, HC=), 5.54 (t, J = 7.4 Hz, 1H, HC=), 3.94 (brs, 1H, CH), 3.71 (s, 3H, OCH_3), 3.65 (s, 3H, OCH_3), 3.23 (dt, J_1 = 12.5 Hz, J_2 = 7.5 Hz, 1H, CH), 2.83—2.74 (m, 1H, CH), 2.57 (dd, J_1 = 13.8 Hz, J_2 = 8.2 Hz, 1H, one proton of CH_2), 2.27 (dd, J_1 = 13.8 Hz, J_2 = 3.8 Hz, 1H, one proton of CH_2), 2.10 (qd, J_1 = 7.2 Hz, J_2 = 3.1 Hz, 2H, CH_2), 2.03 (dd, J_1 = 13.6 Hz, J_2 = 8.0 Hz, 1H, one proton of CH_2), 1.88 (t, J = 12.8 Hz, 1H, one proton of CH_2), 1.41—1.24 (m, 4H, 2 × CH_2), 0.90 (t, J = 7.0 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 172.9, 172.3, 143.0, 136.6, 131.7, 131.5, 129.6, 128.7, 128.1, 127.4, 59.5, 52.7, 52.6, 45.5, 40.3, 40.1, 38.8, 37.8, 32.1, 27.0, 22.2, 13.9; MS (ESI) *m/z*: 441 ([M(^{37}Cl)+Na]$^+$), 439 ([M(^{35}Cl)+Na]$^+$), 419 ([M(^{37}Cl)+H]$^+$), 417 ([M(^{35}Cl)+H]$^+$); IR (neat) *v*: 3018, 2951, 2862, 1733, 1486, 1441, 1395, 1248, 1195, 1163, 1102, 1060, 1014 cm^{-1}; HRMS calcd for $C_{24}H_{33}{}^{35}ClNO_4$ ([M(^{35}Cl)+NH_4]$^+$): 434.2093, found: 434.2091.

(8) Preparation of (3a*S*,5*R*,7a*S*,*Z*)-5-methyl-4-(2-phenylethylidene)-2-((4-methylphenyl)sulfonyl)-2,3,3a,4,5,7a-hexahydro-1*H*-isoindole (3a*S*,5*R*,7a*S*,*Z*)-4b (hanyl-10-011)



(S_a,2*E*,4*E*)-**3b** 94% *ee*

(3a*S*,5*R*,7a*S*,*Z*)-**4b** 80%, 94% ee

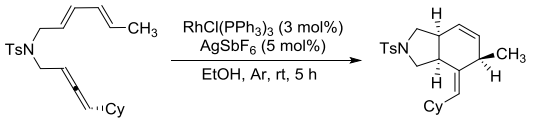
Following **Typical procedure IV**, the reaction of $RhCl(PPh_3)_3$ (13.8 mg, 0.015 mmol), $AgSbF_6$ (8.8 mg, 0.025 mmol), and (S_a,2*E*,4*E*)-**4b** (197.4 mg, 0.5 mmol, 94% *ee*) in EtOH (5.0 mL) afforded (3a*S*,5*R*,7a*S*,*Z*)-**4b** (158.4 mg, 80%) (eluent: petroleum ether/ethyl acetate = 15/1) as a white solid: m.p. 117—119 °C (dichloromethane/pertrum ether), 94% *ee* (HPLC conditions: chiralcel OD-H column, hexane/*i*-PrOH = 95/5, 1.0 mL/min, λ = 214 nm, t_R(minor) = 22.97 min, t_R(major) = 29.65 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.69 (d, J = 8.4 Hz, 2 H, ArH), 7.28 (t, J = 8.2 Hz,

 © 2019 SIOC, CAS, Shanghai, & WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

16147065, 2019, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.201900023 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

4H, ArH), 7.19 (t, *J* = 7.4 Hz, 1H, ArH), 7.09 (d, *J* = 7.2 Hz, 2H, ArH), 5.59 (t, *J* = 7.6 Hz, 1H, HC=), 5.43 (dt, J_1 = 10.0 Hz, J_2 = 3.0 Hz, 1H, HC=), 5.16 (d, *J* = 10.0 Hz, 1H, HC=), 3.48 (dd, J_1 = 10.2 Hz, J_2 = 6.2 Hz, 1H, one proton of NCH_2), 3.41—3.21 (m, 5H, CH_2 of Bn + two protons of NCH_2 + CH), 3.14 (dd, J_1 = 10.4 Hz, J_2 = 9.2 Hz, 1H, one proton of NCH_2), 2.86—2.75 (m, 1H, CH), 2.70—2.62 (m, 1H, CH), 2.41 (s, 3H, CH_3), 1.12 (d, *J* = 7.6 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 143.3, 140.5, 137.2, 133.8, 132.9, 129.5, 128.4, 128.1, 127.5, 127.2, 126.0, 125.8, 54.5, 50.5, 39.2, 38.1, 35.3, 33.0, 24.8, 21.4; MS (ESI) *m*/*z*: 416 ([M+Na]$^+$), 394 ([M+H]$^+$); IR (neat) *v*: 3015, 2954, 2907, 2891, 2865, 1596, 1490, 1451, 1337, 1292, 1229, 1177, 1153, 1130, 1105, 1086, 1052, 1023 cm^{-1}; Anal Calcd for $C_{24}H_{27}NO_2S$: C 73.25, H 6.92, N 3.56. found: C 73.27, H 6.99, N 3.50.

(9) Preparation of (3a*S*,5*R*,7a*S*,*Z*)-5-methyl-2-((4-methylphenyl)-sulfonyl)-4-cyclohexylmethylene-2,3,3a,4,5,7a-hexahydro-1*H*-isoindole (3a*S*,5*R*,7a*S*,*Z*)-4h (hanyl-10-003)



(S_a,2*E*,4*E*)-**3h** 97% *ee*

(3a*S*,5*R*,7a*S*,*Z*)-**4h** 77%, 96% *ee*

Following **Typical procedure I**, the reaction of $RhCl(PPh_3)_3$ (13.8 mg, 0.015 mmol), $AgSbF_6$ (8.7 mg, 0.025 mmol), and (S_a,2*E*,4*E*)-**3h** (193.4 mg, 0.5 mmol, 97% *ee*) in EtOH (5.0 mL) afforded (3a*S*,5*R*,7a*S*,*Z*)-**4h** (148.3 mg, 77%) (eluent: petroleum ether/ethyl acetate = 20/1) as a solid: m.p. 97—99 °C (dichloromethane/petroleum ether); 96% *ee* (HPLC conditions: Chiralcel AS-H column, hexane/*i*-PrOH = 90/10, 1.0 mL/min, λ = 214 nm, t_R(major) = 17.51 min, t_R(minor) = 20.67 min); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.71 (d, *J* = 8.4 Hz, 2 H, ArH), 7.32 (d, *J* = 8.0 Hz, 2H, ArH), 5.38 (dt, J_1 = 10.0 Hz, J_2 = 3.0 Hz, 1H, HC=), 5.22 (d, *J* = 9.2 Hz, 1H, HC=), 5.10 (d, *J* = 10.4 Hz, 1H, HC=), 3.51 (dd, J_1 = 10.4 Hz, J_2 = 6.8 Hz, 1H, one proton of NCH_2), 3.36 (dd, J_1 = 8.8 Hz, J_2 = 7.6 Hz, 1H, one proton of NCH_2), 3.28 (dd, J_1 = 10.2 Hz, J_2 = 1.0 Hz, 1H, one proton of NCH_2), 3.20—3.12 (m, 1H, CH), 3.07 (dd, J_1 = 10.8 Hz, J_2 = 8.8 Hz, 1H, one proton of NCH_2), 2.74—2.64 (m, 1H, CH), 2.63—2.57 (m, 1H, CH), 2.44 (s, 3H, CH_3), 2.16—2.03 (m, 1H, CH of Cy), 1.76—1.60 (m, 3H, CH_2 + one proton of CH_2), 1.56—1.46 (m, 2H, CH_2), 1.32—0.94 (m, 8H, CH_3 + 2 × CH_2 + one proton of CH_2); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 143.2, 135.5, 133.8, 133.7, 133.0, 129.6, 129.4, 127.5, 125.7, 54.5, 50.9, 39.2, 38.4, 36.0, 35.1, 33.9, 33.3, 25.86, 25.80, 25.77, 24.9, 21.4; MS (ESI) *m*/*z*: 408 ([M+Na]$^+$), 386 ([M+H]$^+$); IR (neat) *v*: 2920, 2852, 1746, 1458, 1375, 1259, 1162, 1093, 1022 cm^{-1}; Anal Calcd for $C_{23}H_{31}NO_2S$: C 71.65, H 8.10, N 3.63. found: C 71.69, H 7.92, N 3.37.

Supporting Information

The supporting information for this article is available on the WWW under https://doi.org/10.1002/cjoc.201900023.

Acknowledgement

Financial support from the National Natural Science Foundation of China (No. 21690063) is greatly appreciated. We thank Miss Xinyu Duan in this group for reproducing the results of (3a*R*,5*R*,7a*R*,*Z*)-**4g** and (3a*S*,5*R*,7a*S*,*Z*)-**4h** presented in this study.

References

[1] López, F.; Mascareñas, J. L. [4+2] and [4+3] catalytic cycloadditions of allenes. *Chem. Soc. Rev.* **2014**, *43*, 2904–2915.

[2] (a) Mauleón, P.; Zeldin, R. M.; González, A. Z.; Toste, F. D. Ligand-Controlled Access to [4+2] and [4+3] Cycloadditions in Gold-Catalyzed Reactions of Allene-Dienes. *J. Am. Chem. Soc.* **2009**, *131*, 6348–6349; (b) Alonso, I.; Trillo, B.; López, F.; Montserrat, S.; Ujaque, G.; Castedo, L.; Lledós, A.; Mascareñas, J. L. Gold-Catalyzed [4C+2C] Cycloadditions of Allenedienes, including an Enantioselective Version with New Phosphoramidite-Based Catalysts: Mechanistic Aspects of the Divergence between [4C+3C] and [4C+2C] Pathways. *J. Am. Chem. Soc.* **2009**, *131*, 13020–13030; (c) González, A. Z.; Toste, F. D. Gold(I)-Catalyzed Enantioselective [4+2]-Cycloaddition of Allene-dienes. *Org. Lett.* **2010**, *12*, 200–; (d) Lainhart, B. C.; Alexanian, E. J. Enantioselective Synthesis of cis-Fused Cyclooctanoids *via* Rhodium(I)-Catalyzed [4+2+2] Cycloadditions. *Org. Lett.* **2015**, *17*, 1284–1287.

[3] (a) Wender, P. A.; Jenkins, T. E.; Suzuki, S. Transition Metal-Catalyzed Intramolecular [4+2] Diene-Allene Cycloadditions: A Convenient Synthesis of Angularly Substituted Ring Systems with Provision for Catalyst-Controlled Chemo- and Stereocomplementarity. *J. Am. Chem. Soc.* **1995**, *117*, 1843–1844; (b) Wender, P. A.; Croatt, M. P.; Deschamps, N. M. Metal-Catalyzed [2+2+1] Cycloadditions of 1,3-Dienes, Allenes, and CO. *Angew. Chem. Int. Ed.* **2006**, *45*, 2459–2462.

[4] Han, Y.; Ma, S. Rhodium-catalyzed highly diastereoselective intramolecular [4 + 2] cycloaddition of 1,3-disubstituted allene-1,3-dienes. *Org. Chem. Front.* **2018**, *5*, 2680–2684.

[5] Trost, B. M.; Fandrick, D. R.; Dinh, D. C. Dynamic Kinetic Asymmetric Allylic Alkylations of Allenes. *J. Am. Chem. Soc.* **2005**, *127*, 14186–14187.

[6] Tang, X.; Huang, X.; Cao, T.; Han, Y.; Jiang, X.; Lin, W.; Tang, Y.; Zhang, J.; Yu, Q.; Fu, C.; Ma, S. $CuBr_2$-catalyzed enantioselective routes to highly functionalized and naturally occurring allenes. *Org. Chem. Front.* **2015**, *2*, 688–691.

[7] Gajewski, J. J.; Jurayj, J.; Kimbrough, D. R.; Gande, M. E.; Ganem, B.; Carpenter, K. On the Mechanism of Rearrangement of Chorismic Acid and Related Compounds. *J. Am. Chem. Soc.* **1987**, *109*, 1170–1186.

[8] Fischer, D. F.; Barakat, S.; Xin, Z.-Q.; Weiss, M. E.; Peters, R. The Asymmetric Aza-Claisen Rearrangement: Development of Widely Applicable Pentaphenylferrocenyl Palladacycle Catalysts. *Chem. Eur. J.* **2009**, *15*, 8722–8741.

[9] Schultz, E. E.; Lindsay, V. N. G.; Sarpong, R. Expedient Synthesis of Fused Azepine Derivatives Using a Sequential Rhodium(II)-Catalyzed Cyclopropanation/1-Aza-Cope Rearrangement of Dienyltriazoles. *Angew. Chem. Int. Ed.* **2014**, *53*, 9904–9908.

[10] Oida, S.; Tajima, Y.; Konosu, T.; Nakamura, Y.; Somada, A.; Tanaka, T.; Habuki, S.; Harasaki, T.; Kamai, Y.; Fukuoka, T.; Ohya, S.; Yasuda, H. Synthesis and Antifungal Activities of R-102557 and Related Dioxane-Triazole Derivatives. *Chem. Pharm. Bull.* **2000**, *48*, 694–707.

[11] Wender, P. A.; Glorius, F.; Husfeld, C. O.; Langkopf, E.; Love, J. A. Transition Metal-Catalyzed [5+2] Cycloadditions of Allenes and Vinylcyclopropanes: First Studies of Endo-Exo Selectivity, Chemoselectivity, Relative Stereochemistry, and Chirality Transfer. *J. Am. Chem. Soc.* **1999**, *121*, 5348–5349.

[12] Crystal data for compound (3a*R*,5*S*,7a*R*,*Z*)-**4b**: $C_{24}H_{27}NO_2S$, MW = 393.52, monoclinic, space group *P* 21, final *R* indices [$I > 2\sigma(I)$], R_1 = 0.0504, wR_2 = 0.1137, *R* indices (all data) R_1 = 0.0645, wR_2 = 0.1218, *a* = 12.3481(18) Å, *b* = 6.9284(10) Å, *c* = 13.470(2) Å, α = 90°, β = 114.449(3)°, γ = 90°, *V* = 1049.0(3) Å^3, *T* = 293(2) K, *Z* = 2, reflections collected/unique 6050/3719 (R_{int} = 0.0292), number of observations [> 2σ(*I*)] 3719, parameters: 255. Supplementary crystallographic data have been deposited at the **CCDC 1528212**.

[13] For selected reviews of cycloaddition, see: (a) Lautens, M.; Klute, W.; Tam, W. Transition Metal-Mediated Cycloaddition Reactions. *Chem. Rev.* **1996**, *96*, 49–92; (b) Carruthers, W. Cycloaddition Reactions in Organic Synthesis, Pergamon, Oxford, Oxford, **1990**, pp. 1–208; For representative examples of 1,3-dienes with alkenes, see: (c) O'Mahony, D. J. R.; Belanger, D. B.; Livinghouse, T. On the Counterion Dependence of the Rhodium(I)-Catalysed [4+2] Cycloaddition - A Remarkable Accelerating Effect of the Hexafluoroantimonate Anion.

 © 2019 SIOC, CAS, Shanghai, & WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

16147065, 2019, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.201900023 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Synlett **1998**, 443–445; (d) Sato, Y.; Oonishi, Y.; Mori, M. Novel Rhodium(I)-Catalyzed Cycloisomerization of 1,3-Diene with Alkene in a Tether: Competitive Reaction with [4+2] Cycloaddition. *Organometallics* **2003**, *22*, 30–32; (e) O'Mahony, D. J. R. D.; Belanger, B.; Livinghous e, T. Substrate control of stereoselection in the rhodium(I) catalyzed intramolecular [4+2] cycloaddition reaction. *Org. Biomol. Chem.* **2003**, *1*, 2038–2040; For representative examples of 1,3-dienes with alkynes, see: (f) Wang, B.; Cao, P.; Zhang, X. An efficient Rh-catalyst system for the intramolecular [4+2] and [5+2] cycloaddition reactions. *Tetrahedron Lett.* **2000**, *41*, 8041–8044; (g) Motoda, D.; Kinoshita, H.; Shinokubo, H.; Oshima, K. Phosphane-Free Rhodium Catalyst in an Anionic Micellar System for [4+2] Annulation of Dienynes. *Angew. Chem. Int. E d.* **2004**, *43*, 1860–1862; (h) Fürstner, A.; Stimson, C. C. Two Manifolds for Metal-Catalyzed Intramolecular Diels-Alder Reactions of Unactivated Alkynes. *Angew. Chem. Int. Ed.* **2007**, *46*, 8845–8849; (i) Kim, S. M.; Park, J. H. ; Chung, Y. K. $Au(PPh_3)OPOF_2$-catalyzed intramolecular [4+2] cycloaddition reaction of dienynes. *Chem. Commun.* **2011**, *47*, 6719–6721.

[14] Crystal data for compound (3a*S*,5*R*,7a*S*,*Z*)-**4b**: $C_{24}H_{27}NO_2S$, MW = 393.52, monoclinic, space group *P* 21, final *R* indices [$I > 2\sigma(I)$], R_1 = 0.0607, wR_2 = 0.1400, *R* indices (all data) R_1 = 0.0661, wR_2 = 0.1434, *a* = 12.3767(19) Å, *b* = 6.9406(11) Å, *c* = 13.508(2) Å, α = 90°, β = 114.445(3)°, γ = 90°, *V* = 1056.4(3) Å^3, *T* = 293(2) K, *Z* = 2, reflections collected/unique 5835/3506 (R_{int} = 0.0318), number of observations [$> 2\sigma(I)$] 6492, parameters: 255. Supplementary crystallographic data have been deposited at the **CCDC 1528213**.

[15] Horvath, A.; Benner, J.; Bäeckvall, J.-E. Enantiocontrolled Synthesis of 3-Pyrrolines from α-Amino Allenes. *Eur. J. Org. Chem.* **2004**, 3240–3243.

[16] Ye, J.; Fan, W.; Ma, S. *tert*-Butyldimethylsilyl-Directed Highly Enantioselective Approach to Axially Chiral α-Allenols *Chem. Eur. J.* **2013**, *19*, 716–719.

Manuscript received: January 15, 2019
Manuscript revised: March 8, 2019
Manuscript accepted: March 11, 2019
Accepted manuscript online: March 13, 2019
Version of record online: April 1, 2019

 © 2019 SIOC, CAS, Shanghai, & WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim

16147065, 2019, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cjoc.201900023 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License